



TM545

Integrated safe motion control

Requirements

Training modules:	TM210 - Working with Automation Studio TM415 - Introduction to mapp Axis TM515 - Programming and commissioning safety applications with mapp Safety
Software	Automation Studio ≥ V4.8 Automation Runtime ≥ 4.8x mapp Safety ≥ 5.12 mapp Motion ≥ 5.12 HW upgrade ≥ 2.1.0.0
Hardware	SG4 CPU with POWERLINK V2 interface / interface card SafeLOGIC (X20SL8101, X20SLX806) ACOPOS P3 with SafeMOTION Display module 8EAD0000.000-1

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1 Introduction

Training module "TM545 - Safe motion control with mapp Safety" is designed to become familiar with the SafeMOTION safety functions and demonstrate how they can be used in Automation Studio and SafeDESIGNER.

This training course will explore the relationship between safety-oriented and non-safety-oriented (standard) applications. It will introduce the safety functions and explain how to use them.



Figure 1: mapp Safety offers the full range of functions for the entire hardware portfolio

The safety functions integrated in the drive open up entirely new possibilities for guaranteeing the safety of personnel while maintaining maximum machine availability.



Training module "TM545 – Safe motion control with mapp Safety" is designed to accompany training seminars but does not represent a full documentation. The complete documentation is available in Automation Help.

1.1 Learning objectives

The goal of this training module is to become familiar with the SafeMOTION safety functions and to learn how they are used.

- Participants will learn the principles on which safe integrated motion control operates.
- Participants will learn about the available safety functions and how they are used (STO, SS1, etc.).
- Participants will learn how to add and configure a safe drive in Automation Studio.
- Participants will learn about the relationship between standard and safety applications.
- Participants will learn about the function blocks in the openSAFETY library and the procedure for developing safety functions.
- Participants will learn the procedure for commissioning and maintenance.



In order to correctly implement a safety application, it is important that applicable regulations and standards are observed in all phases of the safety application's lifecycle. This training module only covers the use of the integrated safety functions from SafeMOTION functions in SafeDESIGNER. This training manual can therefore never replace sound training in safety-related topics.

2 Operating principle of "safe motion control"

From the perspective of the standard application, the ACOPOS servo family exhibits the same behavior in the version with and without integrated safety technology. The drive can therefore be integrated in the POWERLINK network as usual and operated in Automation Studio via mapp Motion function blocks.

What makes the ACOPOS servo family with SafeMOTION unique is the additional software and hardware. Here, the encoder signal is evaluated with regard to safety, pulse disabling is checked and the motor holding brake output is controlled.

The purpose of this section is to explain the fundamental characteristics of motion control with integrated safety technology as well as to present its main advantages.



Figure 2: mapp Safety

2.1 Comparison to motion control without integrated safety technology

The fundamental purpose of safe motion control is to interrupt the power supply to the motor. This interruption is performed using safe pulse disabling.

Motion control without SafeMOTION

- Pulse disabling is controlled externally via two inputs

Motion control with SafeMOTION

- **Safe pulse disabling** is controlled by the SafeMOTION module within the module.
- The **safe motor holding brake output** is switched and monitored on the inverter module
- The **safe encoder signal** is evaluated in order to monitor the speed and position limits as needed



SafeMOTION makes it possible to monitor the safe speeds and positions. A moving axis can be used to restore a safe state!

The integrated safety technology does not actively intervene in control processes and therefore only handles safety and monitoring functions.

Stopping an axis

The advantage of integrated safety technology is the possibility to respond to safety-related events. The drive does not necessarily have to be powered off, but can functionally be brought to a standstill in a controlled manner by the standard application and monitored by the SafeMOTION module.

Commands to stop or switch off the axis are already known and are also called from a program on the PLC. The safety application can now set up a time or ramp window and handle standstill monitoring, for example.

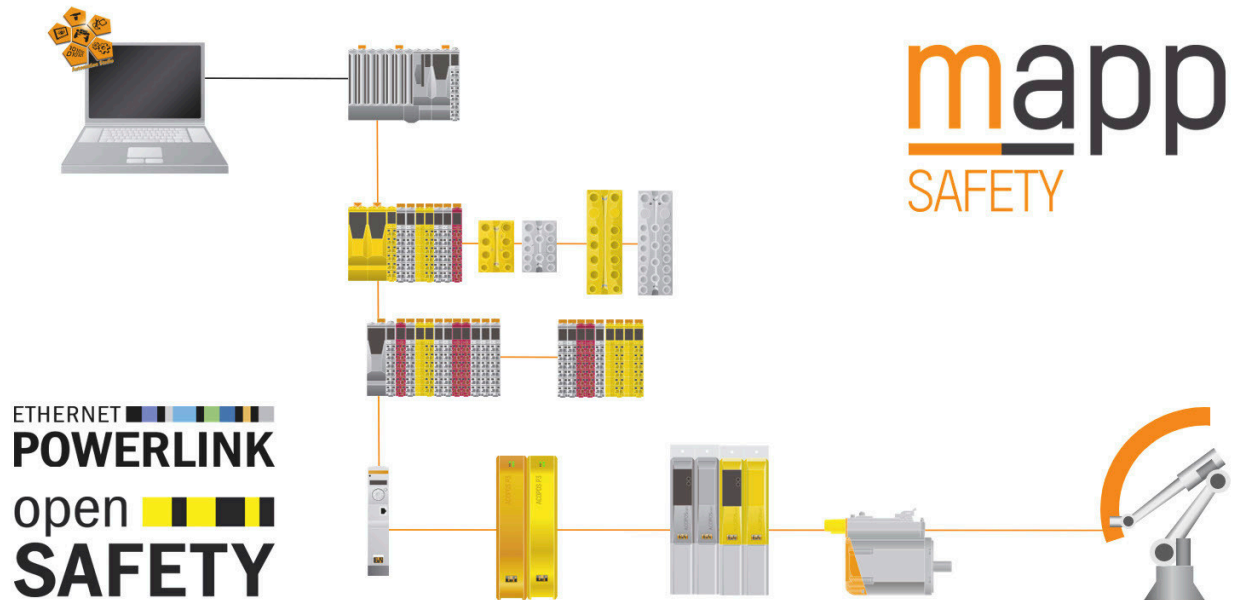


Figure 3: Hardware topology with integrated safety technology

If a safe state of the machine is no longer given and an unauthorized movement is executed, the safety application enables safe pulse disabling of the inverter and the holding brake. This ensures that no additional energy is introduced into the system.

Safety violations should not be the standard case. Safe motion control technology is purely used for monitoring a safe state. Interaction with the standard application is therefore required at all times.

The response times achieved minimize residual movement in the event of error resulting in a significant increase in safety.

2.2 Safe power transmission system

The safe power transmission system consists of the following components:

- SafeMOTION-compatible device
- Encoder cable
- Motor cable
- Motor with position encoder

The safe drive module basically consists of a standard ACOPOS system with additional SafeMOTION hardware and firmware. The encoder interface / SafeMOTION module is permanently installed on the inverter module and cannot be replaced.

Depending on the SafeMOTION safety functions required, a license package is required to enable them.

In the first steps, the drive is commissioned as usual as a familiar ACOPOS device with all configuration settings and test options in Automation Studio. In addition, the SafeLOGIC controller must enable the controller via the "Enable" signal.

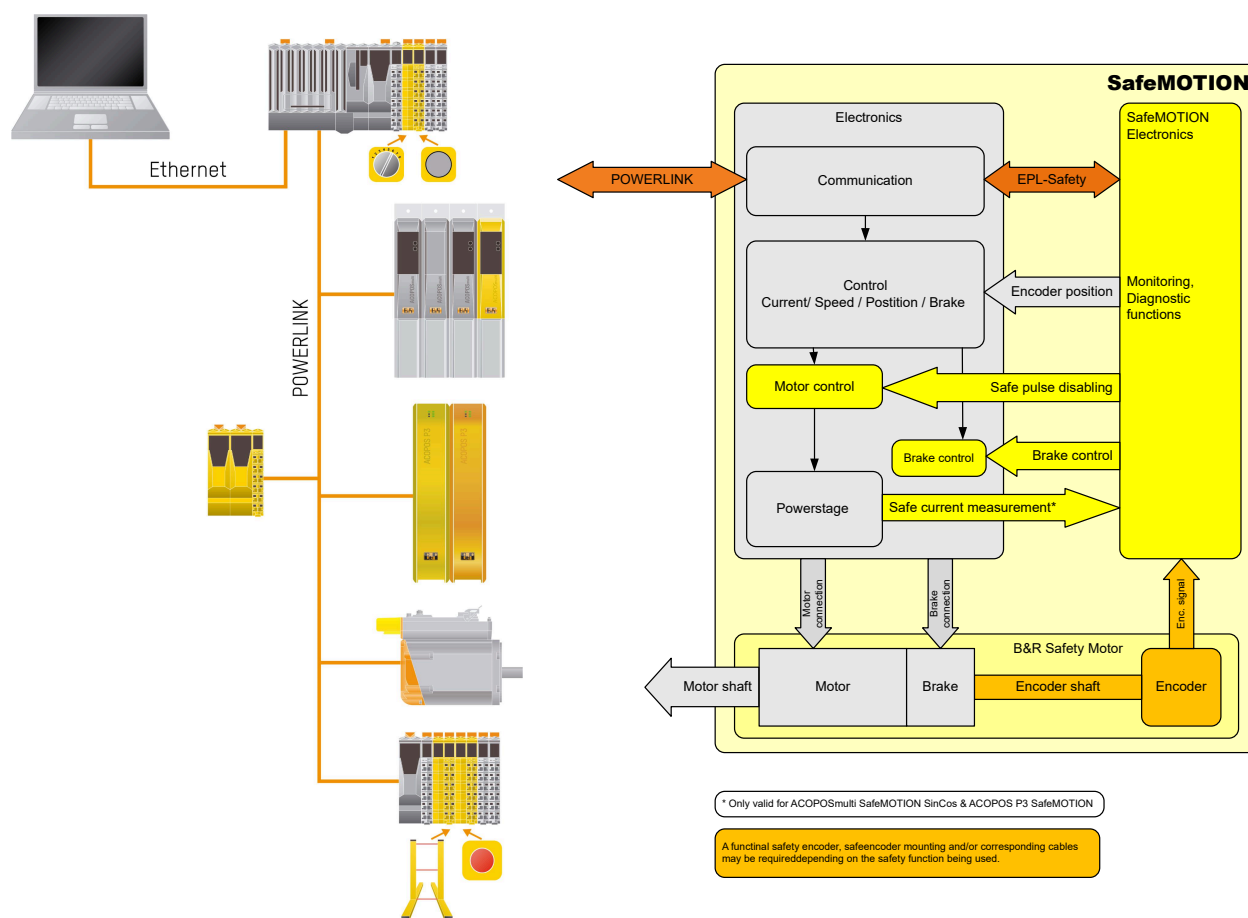


Figure 4: A safe power transmission system demonstrated via an ACOPOSmulti SafeMOTION example



Safety technology \ mapp Safety \

- General information \ Licensing
- Engineering \ SafeMOTION \ The safe power transmission system

2.3 The idle current principle

Integrated safety technology with SafeMOTION uses the idle current principle. When there is a logical 0 at a controller input, the corresponding safety function or error response is executed.

The idle current principle ensures that the system changes to the safe state and safe pulse disabling is enabled. No more energy can be introduced into the system to cause additional movement. If the axis is in motion at the time the error occurs, it will coast to a stop.

In engineering, the generalization of this principle is referred to as **"FAIL SAFE"** (see **"FAIL SAFE state"** on page 9).

This is why cutting off the drive's energy and torque is the only safe function that can be executed at any time.

As a last consequence, safe pulse disabling is always enabled in the event of a safety violation, since this initiates the safe state. The axis is thus switched to a torqueless state, even if a movement is performed.



Situations involving external forces (e.g. suspended loads) can result in dangerous movements! If this poses a safety risk, the user must implement the necessary measures to eliminate the risk (e.g. mechanical brakes)! These measures must correspond to the required safety level!



Safety technology \ Intended use \ Safe state \ The idle current principle

2.4 Pulse disabling and motor holding brake output

SafeMOTION does not actively intervene in open and closed control loops of the inverter module. Only safe pulse disabling and the safe motor holding brake output are operated directly.

Safe pulse disabling

Safe pulse disabling prevents control pulses of the processor from reaching the power unit of the inverter and ensures that the power supply to the motor is interrupted so that it loses all torque/power.

No external wiring is required for SafeMOTION. Pulse disabling is connected by the SafeMOTION module via two channels within the module. The request for the STO safety function is made via the SafeLOGIC controller and is transferred via the openSAFETY protocol.



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \ Safe actuators \ Safe pulse disabling

Safe motor holding brake output

The safe motor holding brake output can enable the motor brake output independently of the active controller on the inverter module. A transistor interrupts the flow of current, the magnetic field in the coil is weakened and the motor brake engages.

The voltage on the motor brake output is evaluated with regard to safety by SafeMOTION, with the transistors tested cyclically.



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \ Safe actuators \ Sensors \ Safe motor holding brake output

2.5 Error states with SafeMOTION

There are essentially two error states, whereby pulse disabling and the safe motor holding brake output are not enabled.

The logbook entry in Automation Studio provides additional information about the pending error.

- [04_Betriebszustände_FailSafe](#)
- [04_Betriebszustände_FunctionalFailSafe](#)

2.5.1 FAIL SAFE state

If a hardware or firmware error occurs, the safe inverter module switches to the non-acknowledgeable error state FAIL SAFE.

A logbook entry in Automation Studio provides more detailed information regarding a pending error, which can also be evaluated in the standard application. In addition, the FAIL SAFE state is indicated by the corresponding LEDs on the device used.

The error may have been triggered by defective hardware or a configuration mistake.

The following behavior results from the FAIL SAFE state:

- Safe pulse disabling is always active, i.e. the motor is no longer supplied with power and does not generate torque.
- In this state, the motor holding brake output is always set to 0 V and the motor holding brake is thus engaged. If the drive was in motion beforehand, the motor holding brake will suffer mechanical wear.

To return the drive to the OPERATIONAL state, a power off / power on cycle must be performed.



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \ Integrated safety functions \ State FAIL SAFE

2.5.2 FUNCTIONAL FAIL SAFE state

If a monitored limit is exceeded or an encoder error occurs during operation, then the device changes to an acknowledgeable error state – the FUNCTIONAL FAIL SAFE state – as long as safe evaluation of the encoder signal is required for the safety functions being used. Information about any errors that occur can be found in the logbook entry in Automation Studio.



If the module switches to the FUNCTIONAL FAIL SAFE state:

- The **S_NotErrFUNC** output on the function block is reset.
- The drive loses all torque/power and coasts to a stop!
- The motor holding brake output is set to 0 V, which engages a connected motor holding brake.
- In the event of an error, a synchronous axis will no longer be synchronous.

If cutting off torque and coasting to a stop on the machine is a problem, an STO1 with delayed STO can also be set in the SafeMOTION parameters. This provides the possibility for the standard application to initiate a short circuit stop with the motor. This results in increased braking of the motor – and therefore the axis – using values that go beyond the defined limits.

The temperature of the motor windings will increase, and this should be taken into account. If there is a risk of motor overheating, the ACOPOS drive automatically switches off and the motor coasts to a stop.

As an alternative, some type of external brake could be installed on the mechanical system.

3 Configuration

For the realization of the application and analysis of the different behavior patterns of the SafeMOTION functions, the following hardware setup serves as an exercise:

- X20 controller with I/Os
- SL8101 with SafeI/Os
 - Light curtains
 - Acknowledgement button
 - Mode selector switch
 - Emergency stop switch
- POWERLINK bus controller with DI module
- ACOPOS P3 SafeMOTION with two axes

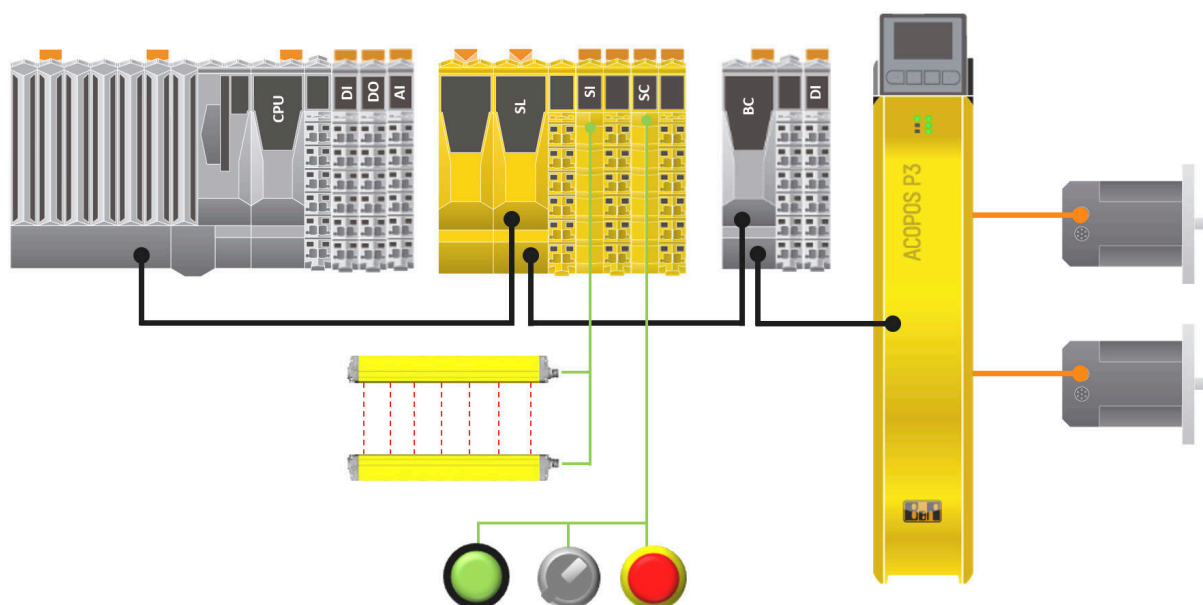


Figure 5: Training hardware

Module	Channel	Function	SafeDESIGNER configuration
X20SI2100	SafeTwoChannelInput0102 - SI_Lightcurtain	Light curtains	Pulse Source = No Pulse Filter off = 1000 µs
X20SC0806	SafeTwoChannelInput0102 - SI_EStop	Emergency stop	
X20SC0806	SafeDigitalInput03 - SI_Automatic	Automatic mode	Pulse source = Pulse 3
X20SC0806	SafeDigitalInput04 - SI_Manual	Manual mode	Pulse source = Pulse 3
X20SC0806	SafeDigitalInput05 - SI_Reset	Green button	Pulse source = Pulse 4

Table 1: Cable pinout

Configuration and programming are split into two areas:

Automation Studio

- Configuring hardware
- Controlling an axis with mapp Motion

SafeDESIGNER

- Configuring safe modules and axes
- Using function block "openSAFETY_BuR_Motion_SF"

3.1 Safety functions in the sample application

The following safety functions must be implemented in sample project "Saw":

Emergency stop

The following function must be implemented:

- Functionally stop the movement when triggered.
- Enable STO with a delay.
- Acknowledge by pressing the green button after startup.
- Acknowledge by pressing the green button after resetting the emergency stop function.
- The simultaneity for multi-channel evaluation must be within **200 ms**.



Figure 6: Emergency stop switch

Light curtains

The following function must be implemented:

- 3 seconds after triggering the light curtain, switch the axis to a torqueless state.
- After startup and triggering the light curtain, acknowledge by pressing the green button.
- The simultaneity for multi-channel evaluation must be within **300 ms**.



Figure 7: Light curtains

Mode selector switch

The following function must be implemented:

- Stop the movement when switching.
- 3 switch positions are evaluated:
 - Manual - Switch position right
 - Automatic - Switch position left
 - Neutral - Switch position middle



Figure 8: Mode selector switch - Switch position right

Green button

The following function must be implemented:

- Start a movement depending on the selected mode of operation.
- Acknowledge the error in the standard application as well as in the safety application.

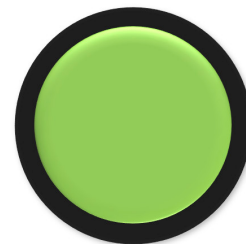


Figure 9: Green button

Operating modes

Automatic operation (switch position left)

- The positive edge of the green button starts a movement with a constant velocity
- The light curtain is active when triggered: Axis stops, SS1 is immediately active, STO after 3 seconds
- A violation of the safety equipment must be acknowledged via a positive edge of the green button.

Manual operation (switch position right)

- As long as the green button is pressed, the axis is in motion
- If the light curtain is triggered, the speed is reduced and monitored with SLS
- Standstill of the axis is monitored with SS2
- Movement monitored with SLI

Neutral operation (Switch position middle)

- Axis is moved using mapp Cockpit
- Acceleration and braking procedure is monitored with SLA according to specified acceleration or deceleration limit values

3.2 Configuration in Automation Studio

At the beginning, all hardware is configured in the Physical View in Automation Studio. The necessary mapp configuration packages are added to the Configuration View. Subsequently, the SafeMOTION axis is controlled via function block MpAxisBasic.

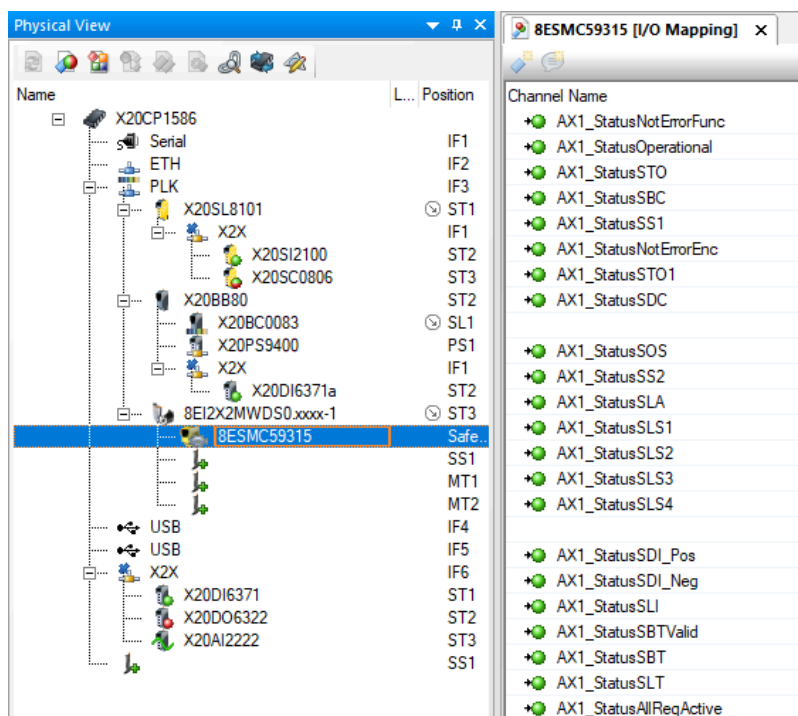


Figure 10: I/O mapping for the SafeMOTION module

The difference for a device with SafeMOTION is that a safe node is additionally added to the Physical View. The status of the various safety functions is available here via the I/O mapping.

Management in Automation Studio includes the following:

- Adding hardware to the Physical View
- Configuring mapp Safety and mapp Motion
- Assigning the axis object to the inverter module
- Controlling an axis in the standard application



Motion control \ mapp Motion
Safety technology \ mapp Safety

Exercise: Configuration and implementation in Automation Studio

The aim of this exercise is to configure the hardware, add the mapp configurations and control the axis in the standard application.

- 1) Create an Automation Studio project.
- 2) Configure the controller, the SafeLOGIC, the I/Os and the ACOPOS drive.
- 3) Add the mapp Safety and mapp Motion configurations.
- 4) Add the ST program, move it to the 10 ms task class and implement function block MpAxisBasic.
- 5) Compile and transfer the project.

3.2.1 Adding the hardware

The hardware modules required are added to the Physical View and connected as needed via POWERLINK or X2X Link.

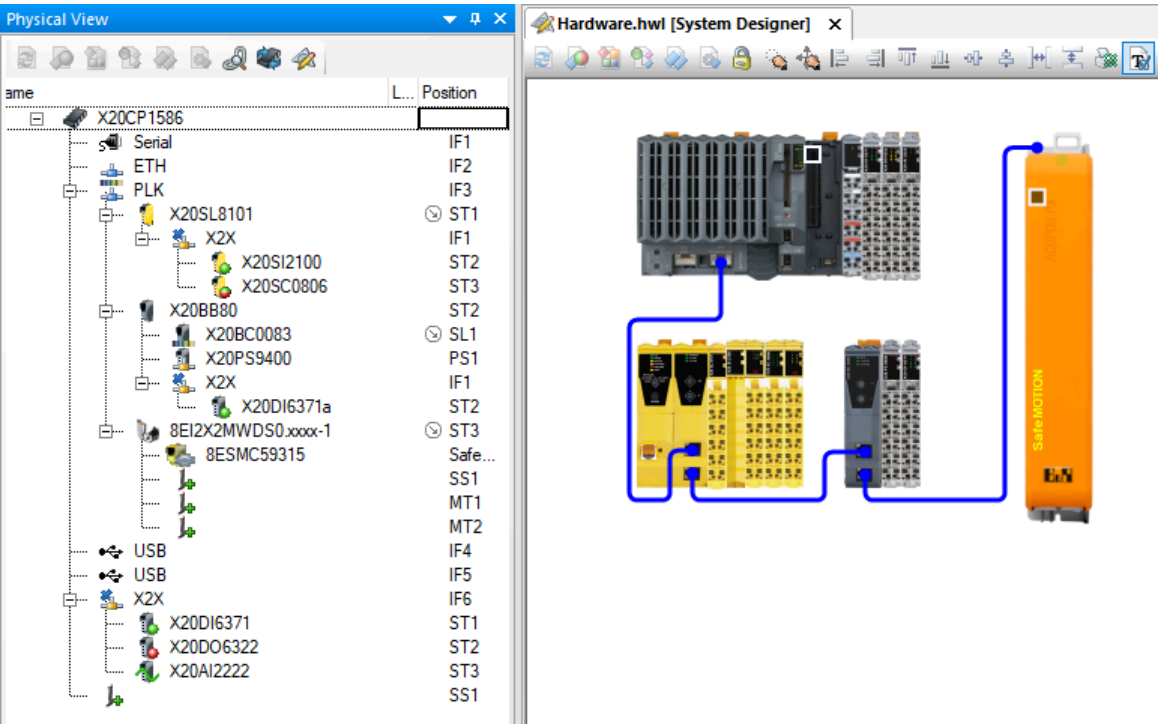


Figure 11: Physical View & System Designer

The configuration of a SafeMOTION axis does not differ from devices without integrated safety technology. However, a SafeLOGIC or SafeLOGIC-X controller is required; otherwise, controller enable cannot be controlled.

The node number on the ACOPOS drive, which must be set via the display module (see image), must match the node number in Automation Studio.

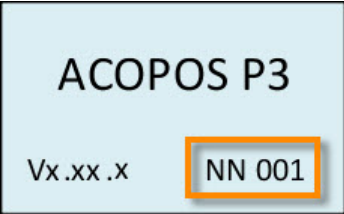


Figure 12: Display module ACOPOS P3 8EAD0000.000-1

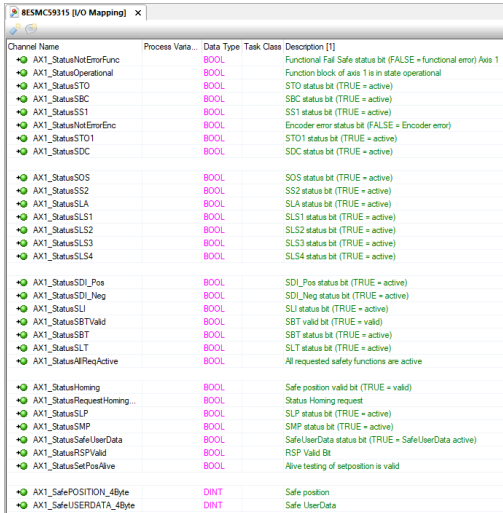


Hardware \ Motion control \ ACOPOS P3 \ Technical data \ Accessories \ 8EAD display modules \ Operation

I/O mapping

The status of individual safety functions can be read out via the I/O mapping for the SafeMOTION module.

These are available in the form of status bits and can be evaluated with the connection to a process variable in the application in Automation Studio.



Channel Name	Process Variable	Data Type	Task Class	Description [1]
AX1_StatusNotErrorFunc		BOOL		Functional Fail Safe status bit (FALSE = functional error) Axis 1
AX1_StatusOperational		BOOL		Function block of axis 1 is in state operational
AX1_StatusSTO		BOOL		STO status bit (TRUE = active)
AX1_StatusSBC		BOOL		SBC status bit (TRUE = active)
AX1_StatusSSI		BOOL		SSI status bit (TRUE = active)
AX1_StatusNotErrorEnc		BOOL		Encoder error status bit (FALSE = Encoder error)
AX1_StatusSTO1		BOOL		STO1 status bit (TRUE = active)
AX1_StatusSDC		BOOL		SDC status bit (TRUE = active)
AX1_StatusSOS		BOOL		SOS status bit (TRUE = active)
AX1_StatusSS2		BOOL		SS2 status bit (TRUE = active)
AX1_StatusSLA		BOOL		SLA status bit (TRUE = active)
AX1_StatusSL1		BOOL		SL1 status bit (TRUE = active)
AX1_StatusSL2		BOOL		SL2 status bit (TRUE = active)
AX1_StatusSL3		BOOL		SL3 status bit (TRUE = active)
AX1_StatusSL4		BOOL		SL4 status bit (TRUE = active)
AX1_StatusSDI_Pos		BOOL		SDI_Pos status bit (TRUE = active)
AX1_StatusSDI_Neg		BOOL		SDI_Neg status bit (TRUE = active)
AX1_StatusSLI		BOOL		SLI status bit (TRUE = active)
AX1_StatusSBTValid		BOOL		SBT valid bit (TRUE = valid)
AX1_StatusSBT		BOOL		SBT status bit (TRUE = active)
AX1_StatusSLT		BOOL		SLT status bit (TRUE = active)
AX1_StatusAllReqActive		BOOL		All requested safety functions are active
AX1_StatusHoming		BOOL		Safe position valid bit (TRUE = valid)
AX1_StatusRequestHoming...		BOOL		Status Homing request
AX1_StatusSLP		BOOL		SLP status bit (TRUE = active)
AX1_StatusSMP		BOOL		SMP status bit (TRUE = active)
AX1_StatusSafeUserData		BOOL		SafeUserData status bit (TRUE = SafeUserData active)
AX1_StatusRSPValid		BOOL		RSP Valid Bit
AX1_StatusSetPosAlive		BOOL		Alive testing of setposition is valid
AX1_SafePOSITION_4Byte		DINT		Safe position
AX1_SafeUSERDATA_4Byte		DINT		Safe UserData

Figure 13: I/O mapping for the SafeMOTION module

3.2.2 Configuration of mapp components

mapp Safety

The configuration for the SafeDOMAIN (Config.sfdomain) is added to the mapp Safety folder in the Configuration View. The name and password are defined for the SafeDESIGNER project.



Figure 14: mapp Safety configuration

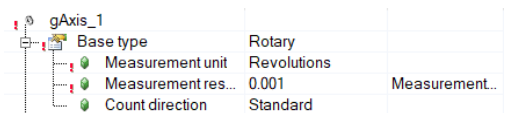
mapp Motion

To control an axis in Automation Studio, a single axis configuration is required in the mapp Motion folder (Config_1.axis). The name of the axis object is defined in this file (gAxis_1).



Figure 15: mapp Motion configuration

It is important to ensure that the same system of units is parameterized for mapp Motion as is used in SafeDESIGNER; otherwise, it is difficult to perform trace analyses.



gAxis_1	Base type	Measurement unit	Measurement resolution	Count direction
	Rotary	Revolutions	0.001	Standard

Figure 16: mapp Motion system of units

To ensure that the axis object and the hardware are connected and the power supply is correctly defined, the configuration of the ACOPOS drive is opened in the Physical View.

The following settings must be made:

- Set the power supply to "ETA system (for training only)".
- Select the axis object that has been added under "Axis reference".

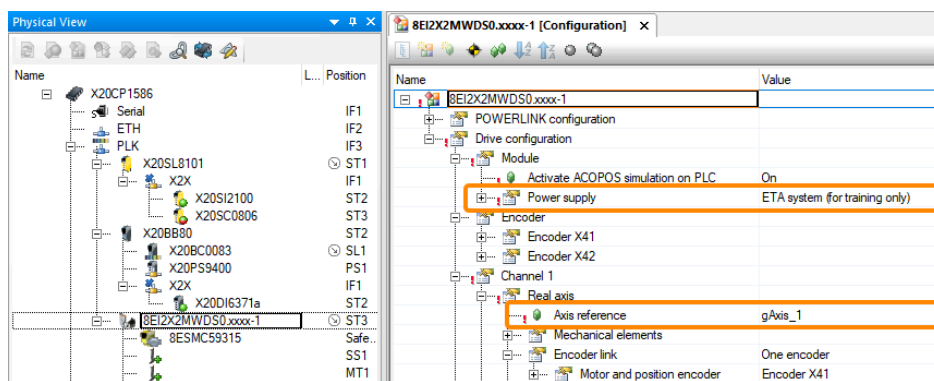


Figure 17: ACOPOS configuration

Select Runtime versions + OPC UA

When using SafeMOTION, the appropriate versions of the following Technology Packages are required:

- mapp View
- mapp Motion
- mapp Safety

Component	Preferred	In use	Scope
Automation Runtime			
Visual Components			
mapp Services	not defined	not defined	
mapp View	not defined	not defined	
ACP10 ARNC0 (Motion)	not defined	not defined	
mapp Motion			
mapp Safety			

Figure 18: mapp View version

Enabling the OPC UA server

If mapp View is used in a project, the OPC UA server must be enabled.

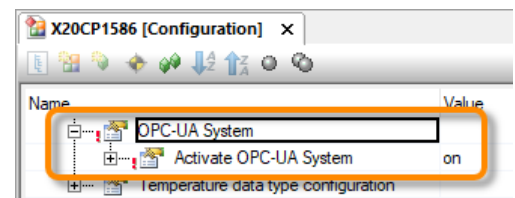


Figure 19: mapp View version

3.2.3 Implementing MpAxisBasic

Function block MpAxisBasic is used to control the axis and read out the status during runtime.

To do this, an ST program is added to the Logical View, an instance of the block is created and the following inputs are connected:

- MpLink → Address of the axis object
- Enable → TRUE
- Parameters → Parameter structure (variable with data type "MpAxisBasicParType")

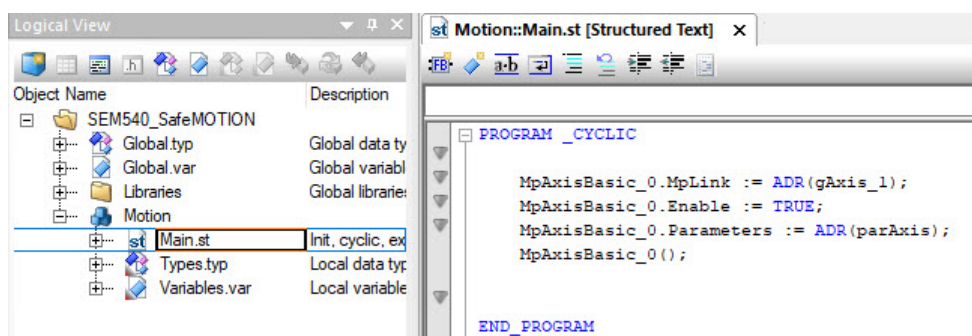


Figure 20: Implementing mpAxisBasic

To enable a faster execution of the program, the task is moved to task class 1 [10 ms].

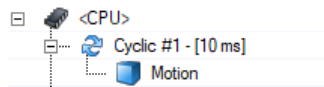


Figure 21: Task class 1

Building and transferring

All changes can now be compiled and transferred to the controller.



Do not forget to enable or disable the SNMP server and the correct IP settings!



Motion control / mapp Motion / Guides / "Getting started" / Axis

3.3 Use in SafeDESIGNER

The entire safety application is created and the individual modules are parameterized in the SafeDESIGNER project. Only devices with SafeMOTION and safety-related components appear in the Safety View. The content of the parameter list on the right side changes with the selected safe module.

ID	Name
SafeDOMAIN 1	
SafeNODE 1 (SL1.SM1)	X20SL8101
SafeNODE 2 (SL1.SM2)	X20SI2100
SafeNODE 3 (SL1.SM3)	X20SC0806
SafeNODE 4 (SL1.SM4)	8ESMC59315

Order no.:	X20SL8101	
Description:	X20 SafeLOGIC, POWERLINK V2, 24V, univ.	
SafeNODE ID:	1	
Import file:	-	

Parameter	Value	Unit
Basic		
Min. required firmware revision	Basic release	
Asynchronous communication load	Auto	
Node guarding timeout	60	s
Auto-acknowledge SafeKEY exchange	No	
Process data transfer rate	High	
Availability source	SafeDESIGNER	
Safety response time default values		
Default safe data duration	20000	us
Default additional tolerated packet loss	1	Packets
Default node guarding packets	5	Packets
Module configuration		

Figure 22: Safety View and parameter list in SafeDESIGNER

Safety applications can be easily created using openSAFETY function blocks. These function blocks are already certified and thus reduce the time required in all phases of the safety application lifecycle.



Safety technology \ mapp Safety \ Programming \ SafeDESIGNER libraries \ openSAFETY_BuR_Motion_SF

Exercise: Parameterization and programming in SafeDESIGNER

The objective of this exercise is to parameterize the axis correctly and to use the openSAFETY function block so that the axis moves.

- 1) Open SafeDESIGNER.
- 2) Set parameter "EUS - Encoder type" to "Encoder used".
- 3) Add library "openSAFETY_BuR_Motion_SF".

- 4) Add function block "SF_oS_MOTION_BR".
- 5) Connect inputs "Activate", "S_AxisID", "S_Control_Reset", "S_Control_Activate".
- 6) Compile and then transfer using the Remote Control dialog box.
- 7) Start the axis movement in Automation Studio.

3.3.2 Parameterization of the safe axis in SafeDESIGNER

Encoder type

To commission the safe axis, the correct parameters must be set.

Since an EnDat2.2 encoder set up for safety must be used for the training hardware, "Encoder used" must be selected in setting "EUS - Encoder type".

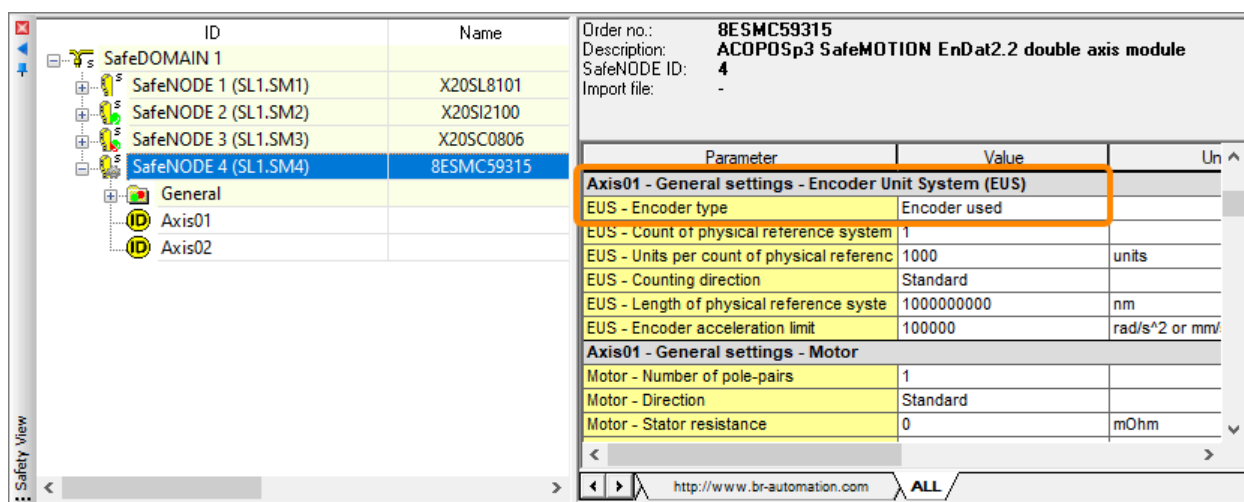


Figure 23: Parameter of the SafeMOTION axis

General settings

The system of units and the direction of rotation of the safe position are defined in this group.

Safety response time

This parameter block concerns the response time of the SafeMOTION. These settings can be defined for all safe nodes in the SafeLOGIC parameters in SafeDESIGNER.

Safe monitoring of the motor/encoder shaft connection

To be able to safely exclude mechanical errors, parameter "Encoder monitoring - Safe encoder mounting" has two setting options:

- **"From motor data record"**
For B&R motors with EnDat2.2 FS encoders, the encoders are mounted strictly according to Heidenhain specifications to avoid mechanical connection errors (breakage, slip, offset). Information about encoder mounting is stored in the motor data, so no further measures are required to monitor the connection between the motor and the encoder shaft. If the encoder is not mounted securely, additional monitoring measures must be enabled.
- **"Approved by user"**
This setting can be used for third-party motors where the encoder is securely attached, but this information is not stored in the motor data.



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \

- Register description SafeMOTION / Parameters in SafeDESIGNER
- Safe actuators \ Sensors \ Safe encoder input \ Mechanical mounting

3.3.3 Adding a library and a function block

Library "openSAFETY_BuR_Motion_SF" is used to implement the SafeMOTION function.

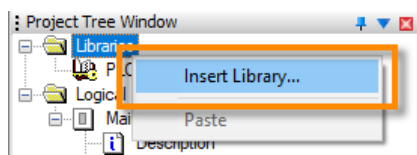


Figure 24: Adding the library to the project tree

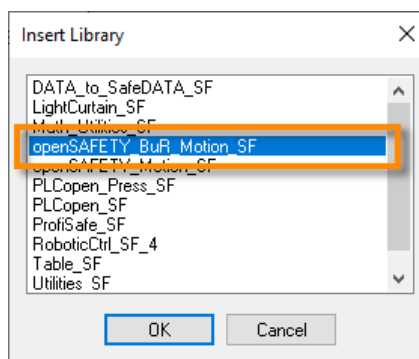


Figure 25: Selecting the library

All function blocks have PLCOpen-specified inputs or outputs:

- Activate → Enabling the function block
- Ready → Function block is executed
- Error → Boolean error message
- DiagCode → Error code

In addition, input "S_AxisID" is used to reference to the axis used.

Function blocks

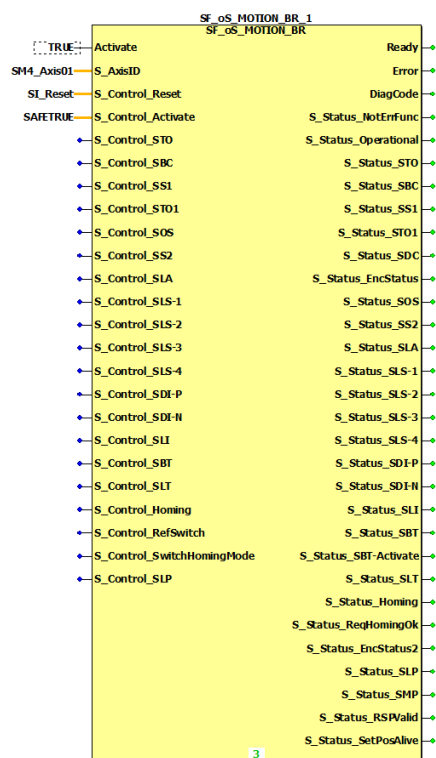


Figure 26: Function block SF_oS_MOTION_BR

Function block "SF_oS_MOTION_BR" is used to map most of the available safety functions.

For clear programming, various safety functions are also grouped into smaller blocks, according to the openSafety drive profile:

- SF_oS_MOTION_Basic_BR
- SF_oS_MOTION_Speed_BR
- SF_oS_MOTION_Advanced_BR
- SF_oS_MOTION_AbsOos_BR

Simultaneous use of an axis with one of the named function blocks and function block SF_oS_MOTION_BR is not permitted

The following function blocks are available to additionally use the cyclic data such as safe speed, safe position or safe torque in the safety application:

- SF_oS_MOTION_Data_Speed_BR
- SF_oS_MOTION_Position_BR
- SF_oS_MOTION_UserData_BR



Connecting the function block

To enable the movement of the axis and to send controller enable from SafeLOGIC to SafeMOTION, function block "SF_oS_MOTION_BR" is added to the worksheet and the following inputs are connected:

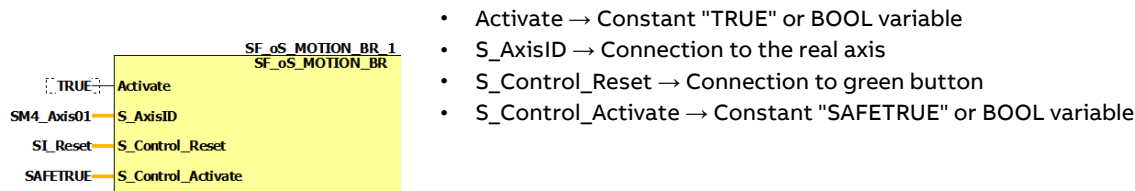


Figure 27: Connecting the inputs

Order no.: X20SC0806
Description: X20 Safe Combined Module, 8x1 24V, 6x0 24V 0.2A
SafeNODE ID: 3
Import file: -

Parameter	Value	Unit
SafeDigitalInout05		
Pulse source	Pulse 4	
Filter off	0	us
Filter on	200000	us
Discrepancy time	50000	us
Dual-channel processing mode	Equivalent	

Figure 28: Pulse source

The pulse source of SafeDigitalInput05 must be set to "Pulse 4" so that the signal of the green button can be read out correctly.

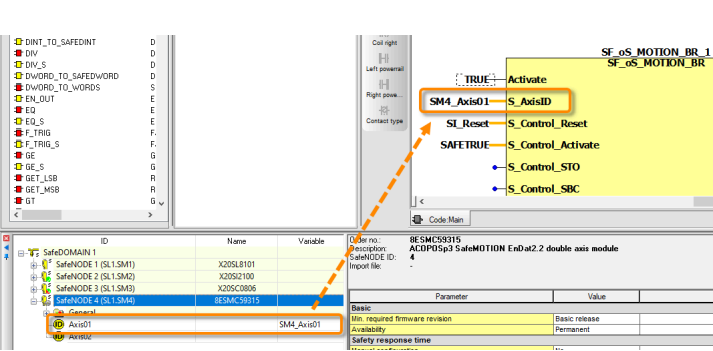


Figure 29: Axis in the Safety View

The connection to the real axis is added via drag-and-drop from the Safety View.

?

Safety technology \ mapp Safety \ Programming \ SafeDESIGNER libraries \ openSAFETY_BuR_Motion_SF\ SF_oS_MOTION_BR

3.3.4 Compiling, transferring and starting an axis

Compiling

The project is compiled before it is transferred to the safety controller.



Figure 30: Compiling

3.3.4.1 Transfer to the SafeLOGIC controller

The safety application can be transferred and commissioned either via the mapp Safety HMI application or the Remote Control dialog box as described in "TM515 - Programming and commissioning a safety application with mapp Safety".

mapp Safety HMI

The following link is used to access the mapp Safety HMI application: <IP address>:81/mappSafety

Connection to SafeDOMAIN ID: 1 X20SL8101 State: Setup Mode Active

After transferring, a reboot of the SafeLOGIC is carried out.

Browse for data Transfer to SafeDOMAIN

☒ SafeAPPLICATION ☐ SafeCOMMISSIONING

Name	Safety
CRC	0xFBC13A34
Time stamp	Thu Jul 16 16:53:54 2020
User	jacob.f.

Format SafeKEY Clear Data

Connection Transfer Completion Exchange

Figure 31: mapp Safety HMI

After compiling the SafeCOMMISSIONING.xml file in SafeDESIGNER, the .swt file is updated in Automation Studio. These changes are now compiled in Automation Studio and transferred to the gray controller. The mapp Safety HMI application is called via a browser and the SafeAPPLICATION is now loaded from the gray controller to the SafeLOGIC controller.



Transfer via the mapp Safety HMI application must be used for the first and last step of commissioning. This ensures that the most current SafeAPPLICATION file is stored on the gray controller.



Safety technology \ mapp Safety \ "Getting started" \ Commissioning the preconfigured HMI application

Remote Control dialog box

The changes to the SafeAPPLICATION are compiled in SafeDESIGNER and transferred directly to SafeLOGIC controller using the Remote Control dialog box.

The setup mode is enabled in this dialog box to be able to use the mapp Safety HMI application.

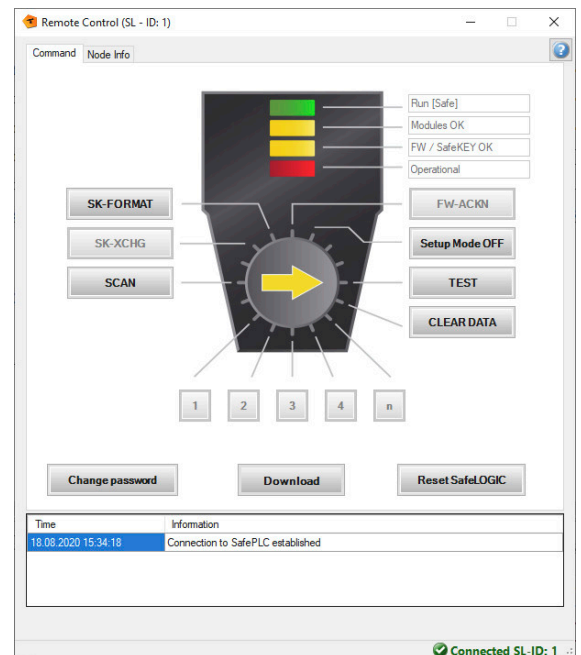


Figure 32: Remote Control dialog box



If the SafeAPPLICATION is transferred via the Remote Control dialog box, only the SafeLOGIC file is updated, but not the file on the gray controller. This mode is recommended only during implementation for testing the SafeAPPLICATION. The user must ensure a complete transfer via the mapp Safety HMI application!



Safety technology \ mapp Safety \ Programming \ SafeDESIGNER \ Dialog boxes for controlling the safety controller \ Remote Control

Starting an axis movement



To be able to perform the first movement, the axis must first be enabled pressing the green button. This is necessary to perform a start reset on the SafeMOTION module and to switch the internal state machine to the "Operational" state.

Only when output "MpAxisBasic_0.Info.ReadyToPowerOn" is set to TRUE, the following inputs can be set one after the other via the monitor mode or the Watch window:

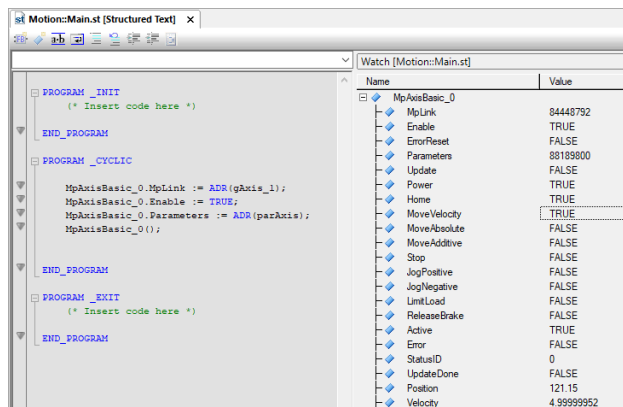


Figure 33: Monitor mode

- 1) Set input "Power" to TRUE.
- 2) Set input "Home" to TRUE.
- 3) Set input "MoveVelocity" to TRUE.

4 Error handling

During development and commissioning, various errors can occur on the drive or in SafeMOTION. Diagnostics possibilities are presented in the following chapter.

4.1 Status indicators on the drive

LED status indicators on the drive



Figure 34: Axis status display module

One method of obtaining the status of the drive is via the LED indicators on the drive itself.

Display module

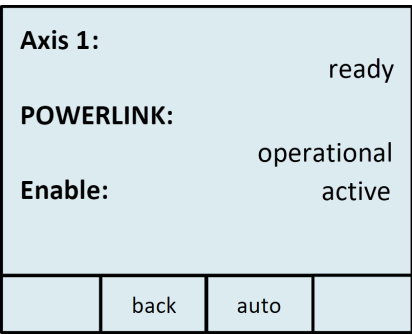


Figure 35: Axis status display module

With the ACOPOS P3, additional information, such as the axis status, can be read out via the display module. Automation Help describes the menu navigation and function of the display module in more detail.



Hardware \ Motion control \ ACOPOS P3 \
..\ Technical data \ 8EI servo drives \ Display
..\ Accessories \ 8EAD display module \ 8EAD0000.000-1 display module \ Operation

4.2 Logger

During development, the module may start with the FAIL SAFE state or switch to the FUNCTIONAL FAIL SAFE state unexpectedly. The most common reason for this is an incorrect parameterization in SafeDESIGNER. The Logger can be used to identify the cause of the error.

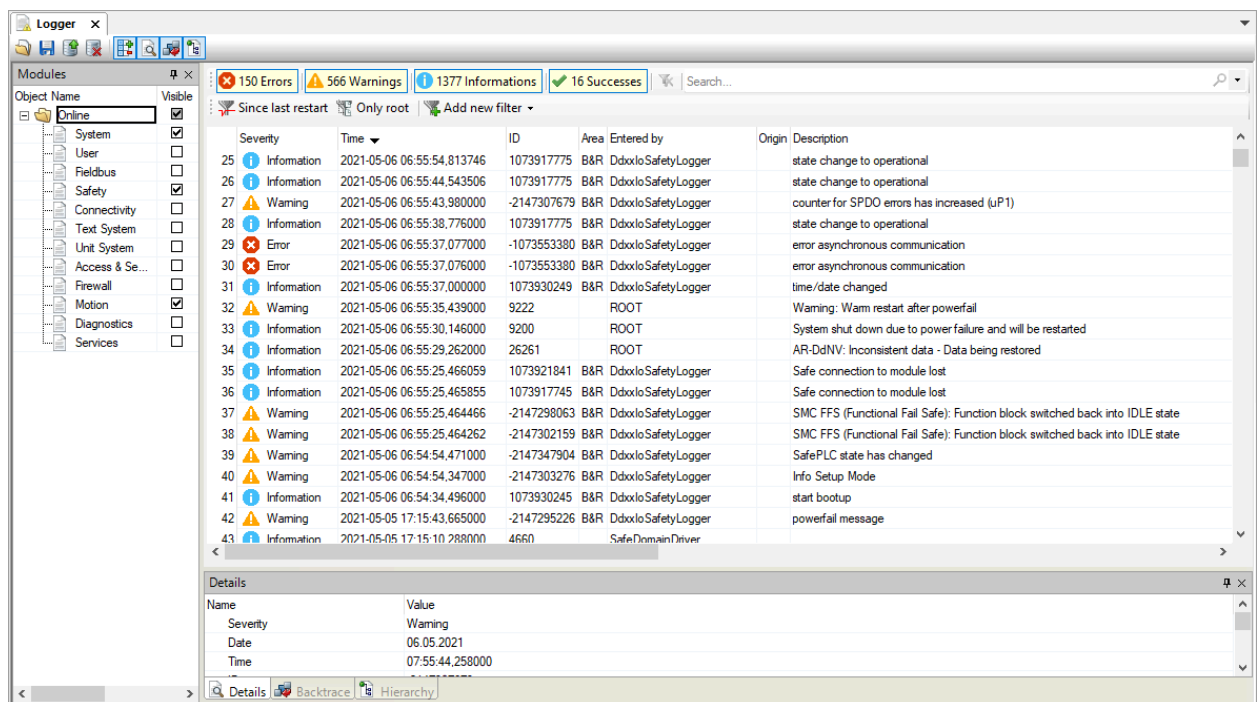


Figure 36: Logger

The short description of the Logger only provides basic information. The complete error text and details are available in Automation Help.



Diagnostics and service \ Diagnostics tools \ Logger

4.3 mapp Cockpit

The web-based mapp Cockpit HMI application can be reached using Google Chrome via the following URL: <IP address>:8084/mappCockpit

There is a wide range of functions that are helpful when commissioning or troubleshooting a drive axis or axis group:

- Testing the behavior by executing a command
- Monitoring the behavior (observing live values, recording a trace and checking logged events)
- Changing the configuration to achieve the desired behavior and, if necessary, testing the behavior once again

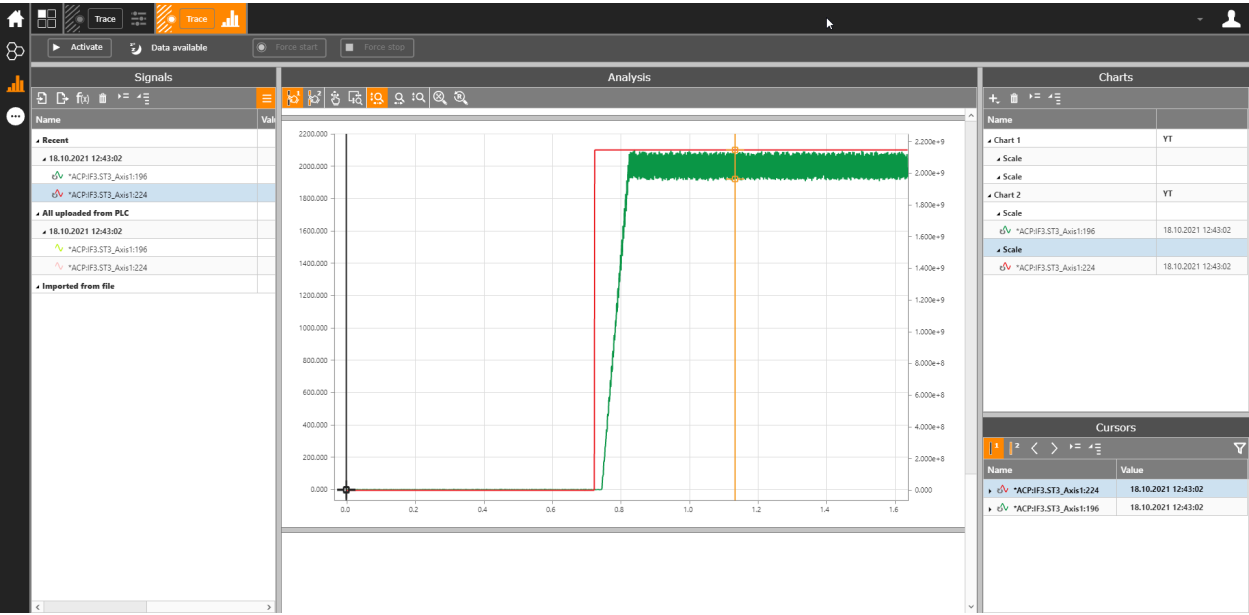


Figure 37: mapp Cockpit

A "Getting started" tutorial in mapp Cockpit as well as detailed information are available in Automation Help and in training module "TM415 - Basic information mapp Axis ".



Diagnostics and service \ mapp Cockpit

5 Safety functions and their application

The following section describes selected integrated safety functions and how they are used. A list of all available safety functions and their safety level are available in Automation Help:



Safety technology \ mapp Safety \ Engineering
 Safety level overview for safety functions \
 Safety level overview for ACOPOS product family safety functions\
 SafeMOTION integrated safety technology ("network-based safety technology")

The following applies to all monitoring safety functions:

If a current speed or position limit is violated, then SafeMOTION switches to the acknowledgeable FUNCTIONAL FAIL SAFE error state.

Detailed information is available under "[FUNCTIONAL FAIL SAFE state](#)" on page

5.1 Safe Torque Off (STO)

STO is the fundamental safety function for SafeMOTION since it represents the implementation of the idle current principle.

A request from the STO enables safe pulse disabling and switches off the torque and power to the drive. Safe pulse disabling is enabled via a signal that is sent directly to the ACOPOS drive via the openSAFETY protocol.

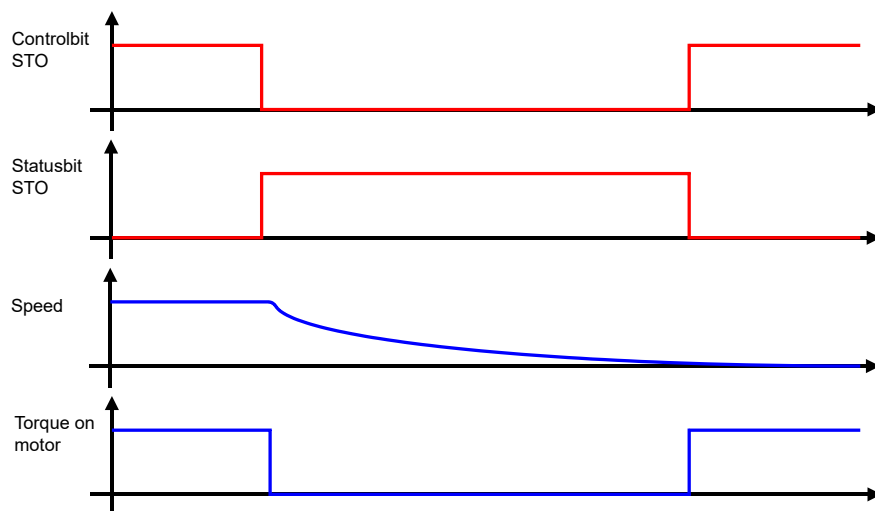


Figure 38: Safe Torque Off, STO



A STO request causes synchronized axes to no longer be synchronous.
 If the drive is in motion at the time STO is requested, it will coast to a stop.
 The resulting residual movement and time depends on the properties of the machine and must always be considered when dimensioning the safety equipment.
 The maximum possible movement (worst case) must be assumed!

Application of the safety function

When STO is triggered, the drive is immediately prevented from supplying torque-generating power. This function can be used for motors that are able to reach standstill in a sufficiently short amount of time on their own (e.g. due to friction) or where coasting to a standstill is irrelevant from a safety point of view.

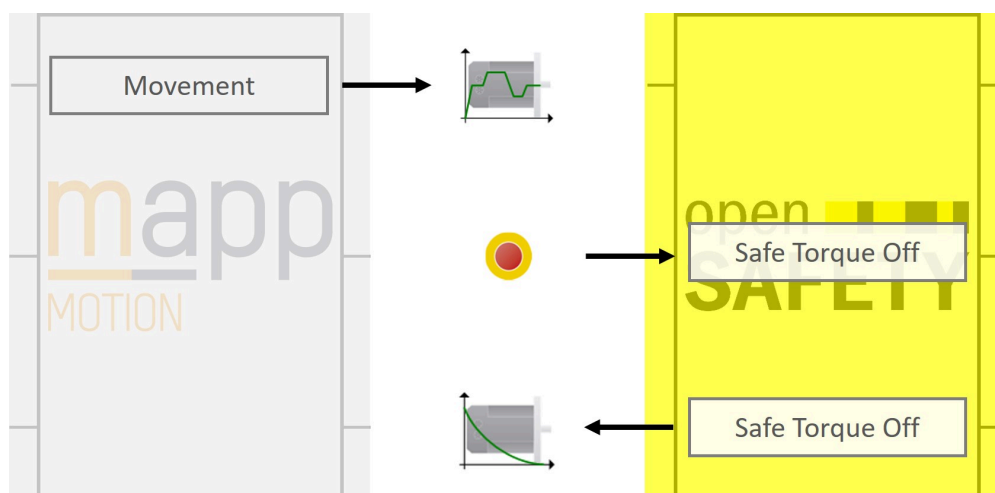


Figure 39: Safe Torque Off, STO



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \ Integrated safety functions \ Safe Torque Off, STO

Exercise: Emergency stop function

The aim of this exercise is to stop the movement via the emergency stop button and to enable pulse disabling using the STO safety function.

SafeDESIGNER:

- 1) Add and connect SF_EmergencyStop_V1.
- 2) Connect output "S_EStopOut" with input "S_Control_STO".
- 3) Compiling

Automation Studio:

- 1) Connect the emergency stop signal with global variable "gdiEStop".
- 2) Switch off input "Power" when the emergency stop button is pressed.
- 3) Compiling

Transferring the SafeAPPLICATION

Testing:

- 1) Start the axis movement in monitor mode.
- 2) Press the emergency stop button → The axis is stopped.
- 3) Check DriveEnable of output mpAxisBasic.
- 4) Check the axis status on the ACOPOS display module.
- 5) Acknowledge function block "SF_EmergencyStop" by pressing the green button.
- 6) Acknowledge the error in the standard application with "ErrorReset".
- 7) Restart the movement.

```

PROGRAM _INIT
    MpAxisBasic_0.Enable := TRUE;
END_PROGRAM

PROGRAM _CYCLIC
    MpAxisBasic_0.MpLink := ADR(gAxis_1);
    MpAxisBasic_0.Parameters := ADR(parAxis);

    IF (gdiEStop = FALSE) THEN
        MpAxisBasic_0.Power := FALSE;
    END_IF;

    MpAxisBasic_0();
END_PROGRAM

```

Table 2: Example code for the emergency stop function - Automation Studio

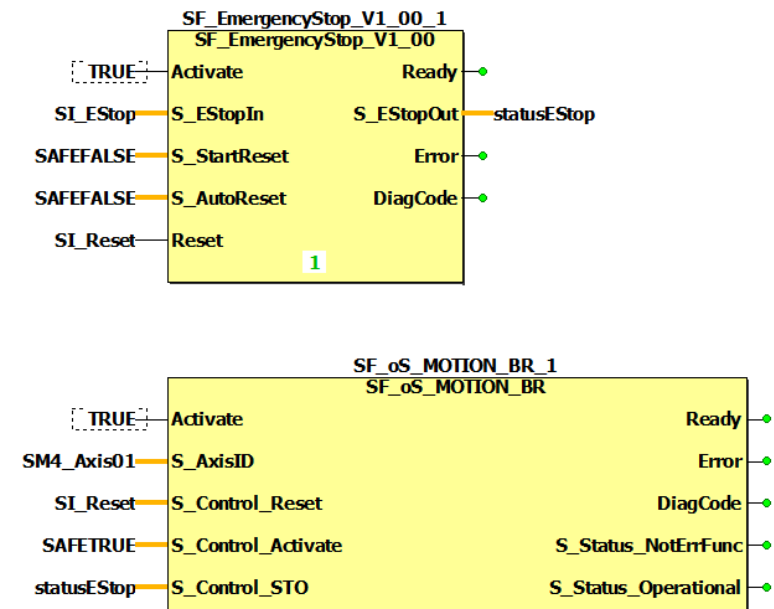


Figure 40: Example code for the emergency stop function - SafeDESIGNER

5.2
Safe Stop 1 (SS1)

When requesting the SS1 safety function, the deceleration process of the axis is monitored until standstill after ramp delay time "SS1 - Ramp monitoring - Time" has passed. After decelerating, safe pulse disabling is enabled and switches off the torque/power to the drive.

If a violation occurs during ramp monitoring, safe pulse disabling is enabled immediately and the drive switches to acknowledgeable error state FUNCTIONAL FAIL SAFE.

Deceleration is controlled by the non-safety-related standard application.

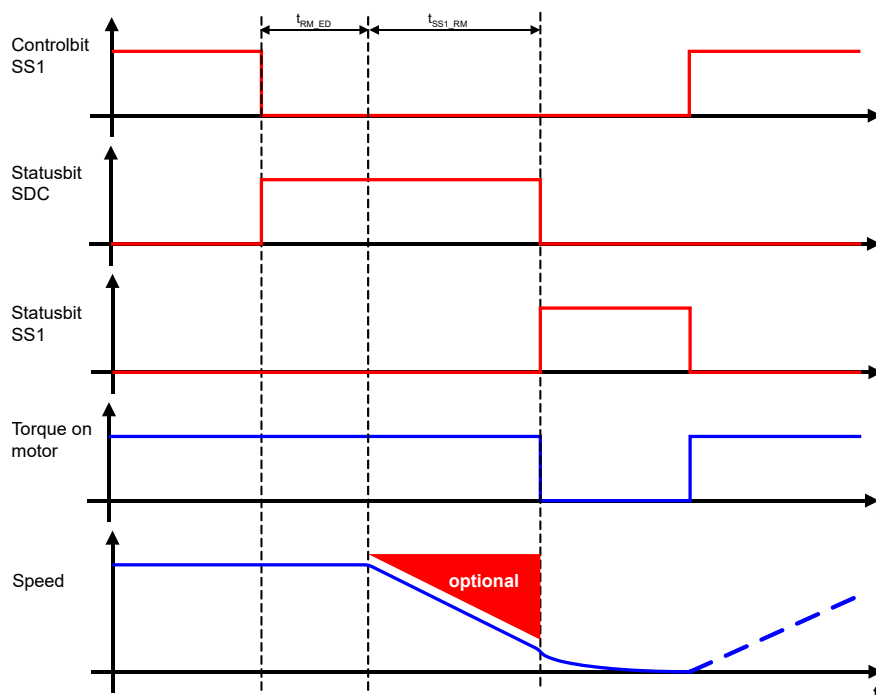


Figure 41: Safe Stop 1

The monitored ramp always starts at the currently monitored limit and is calculated using the parameterized slope. If no speed limit is specified, the monitored ramp starts at the **maximum possible speed $2^{31} - 1 = 2147483647$ [unit/s]**. This does not make sense in normal cases! In order to start the ramp at a physically reasonable value, it is recommended to limit the currently monitored speed limit using safety function Safe Maximum Speed (SMS), for example siehe "Safe Maximum Speed (SMS)" auf Seite 39.

Application of the safety function

The SS1 function enables a controlled stop function for the axes. This allows active deceleration thereby preventing dangers that could otherwise occur due to axes spinning out.

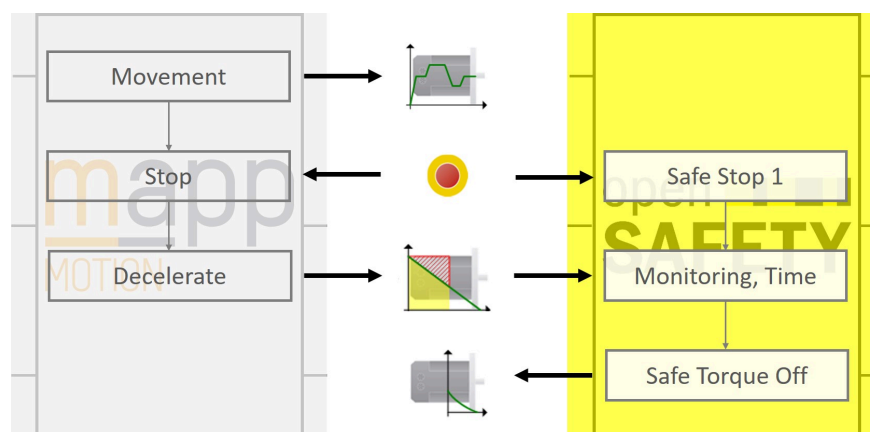


Figure 42: Safe Stop 1



The SS1 safety function does not include monitoring the axis standstill. It simply enables pulse disabling once the axis has been brought to a standstill.



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \ Integrated safety functions \ Safe Stop 1, SS1

Exercise: Safe standstill and pulse disabling (SS1)

The goal of this exercise is to automatically set the axis in motion using the green button, implement the light curtain function and trigger SS1.

Automation Studio:

- 1) Create variables "gdiReset" and "gdiLightcurtain" and connect them with the input.
- 2) Program an automatic procedure using CASE.
- 3) Steps: enINIT, enSTART, enPOWERON, enHOME, enOPERATION, enERROR
- 4) Compiling

SafeDESIGNER:

- 1) Parameterize "SafeDigitalInput01/ 02" with "No Pulse" and "Filter off".
- 2) Add "SF_ESPE_V1_00" and connect it.
- 3) Connect output "S_ESPE_Out" with input "S_Control_SS1".
- 4) Enable SS1 and set "SS1 - Ramp monitoring - Time" to 3 seconds.
- 5) Compiling

Transferring the SafeAPPLICATION**Testing:**

- 1) Start the axis using the green button.
- 2) Trigger the light curtain.
- 3) Check DriveEnable of output mpAxisBasic.
- 4) Check the axis status on the ACOPOS display module.
- 5) Observe the status after 3 seconds.


```

IF gdiEStop = FALSE OR gdiLightcurtain = FALSE THEN
    MpAxisBasic_0.Power := FALSE;
    sStep := enERROR;
END_IF;

CASE sStep OF
    enINIT:
        IF (gdiReset = TRUE) THEN
            sStep := enSTART;
        END_IF;

    enSTART:
        IF (MpAxisBasic_0.Info.ReadyToPowerOn = TRUE
            AND MpAxisBasic_0.Active = TRUE) THEN
            sStep := enPOWERON;
        END_IF;

    enPOWERON:
        MpAxisBasic_0.Power := TRUE;
        IF (MpAxisBasic_0.PowerOn = TRUE) THEN
            sStep := enHOME;
        END_IF;

    enHOME:
        MpAxisBasic_0.Home := TRUE;
        IF (MpAxisBasic_0.IsHomed = TRUE) THEN
            MpAxisBasic_0.Home := FALSE;
            sStep := enOPERATION;
        END_IF;

    enOPERATION:
        MpAxisBasic_0.MoveVelocity := TRUE;
        IF (MpAxisBasic_0.Error = TRUE) THEN
            sStep := enERROR;
        END_IF;

    enERROR:
        MpAxisBasic_0.MoveVelocity := FALSE;
        IF (MpAxisBasic_0.Error = FALSE
            AND MpAxisBasic_0.StatusID = 0) THEN
            IF (MpAxisBasic_0.PowerOn = FALSE
                AND MpAxisBasic_0.Power = TRUE) THEN
                MpAxisBasic_0.Power := FALSE;
            END_IF;
            sStep := enINIT;
        END_IF;
END_CASE;

MpAxisBasic_0.ErrorReset := gdiReset;
MpAxisBasic_0();

```

Table 3: Example code for the light curtain - Automation Studio

Order no.:	8ESMC59315
Description:	ACOPDsp3 SafeMOTION EnDat2.2 double axis module
SafeNODE ID:	4
Import file:	-

Parameter	Value	Unit
Axis01 - General settings - Ramp monitoring		
Ramp monitoring - Speed deceleration limit	1073676289	units/s^2
Ramp monitoring - Enable delay time	100000	us
Axis01 - Basic functions - STO1		
STO1 - Channel	Highside	
Axis01 - Basic functions - SS1		
SS1 - Ramp monitoring - Enable	Enabled	
SS1 - Ramp monitoring - Time	3000000	us

Figure 43: Ramp monitoring parameter - SafeDESIGNER

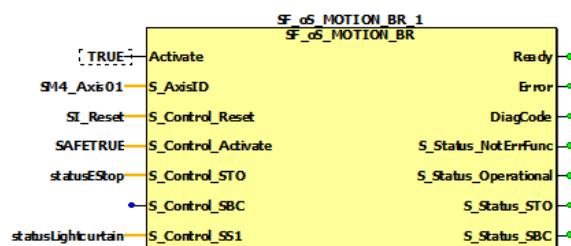
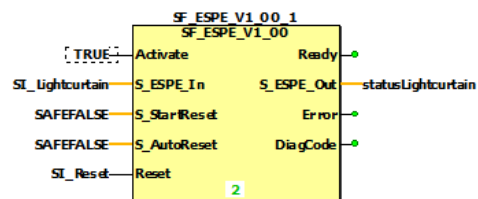


Figure 44: Example code for the light curtain - SafeDESIGNER

5.3 Safe homing

This safety function is used to establish a reference between the encoder position and the machine position.

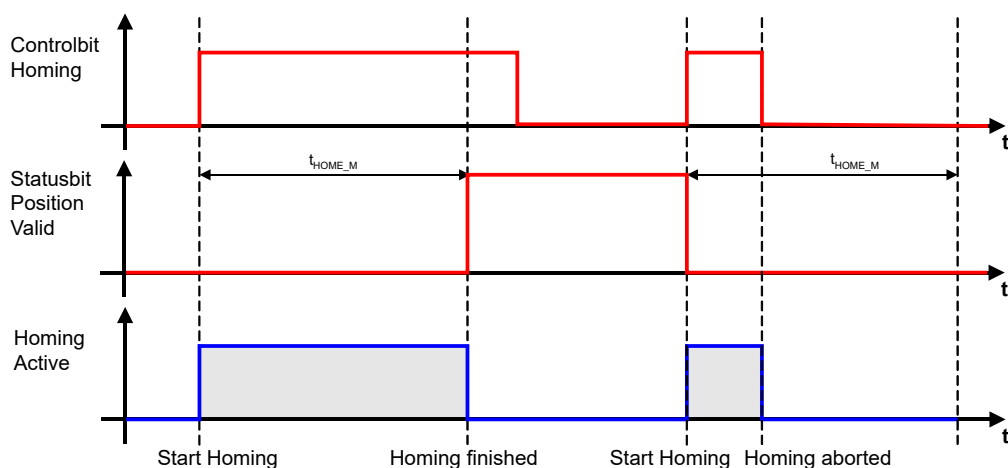


Figure 45: Safe homing

The following safe homing variants are supported:

- Direct
- Reference switch
- Home offset / Home offset with correction (for SafeMOTION EnDat2.2 only)

Safety function "Safe homing" is a prerequisite for implementing safety functions **SLP** and **SMP** and for using the safe position. The SafePositionValid status will remain set to SAFEFALSE until safe homing has been performed!



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \ Integrated safety functions \ Safe homing

Exercise: Safe homing

The goal of this exercise is to directly home an axis in SafeMOTION.

SafeDESIGNER:

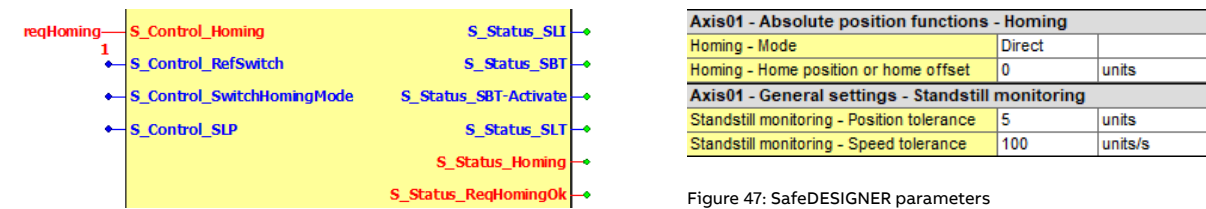
- 1) Define parameter "Homing - Home position".

- 2) Define "Standstill monitoring - Speed tolerance" of 100.
- 3) Define "Standstill monitoring - Position tolerance" of 5.
- 4) Declare local variable "reqHoming".
- 5) Connect the variable with input "S_Control_Homing".
- 6) Compiling

Transferring the SafeAPPLICATION

Test:

- 1) Switch on the axis and home it in Automation Studio.
- 2) Enable the debug mode in SafeDESIGNER.
- 3) Force "reqHoming" to TRUE.
- 4) Check "S_Status_Homing" and "S_Status_ReqHomingOk".



5.4 Safe Stop 2 (SS2)

If the SS2 safety function is enabled, the deceleration process of the axis and standstill are monitored after ramp delay time "SS2 - Ramp monitoring - Time" has passed.

Unlike the SS1 function, the drive must be kept at standstill by the standard application.

If a violation occurs during ramp or standstill monitoring, safe pulse disabling is enabled immediately and the drive switches to an acknowledgeable FUNCTIONAL FAIL SAFE error state.

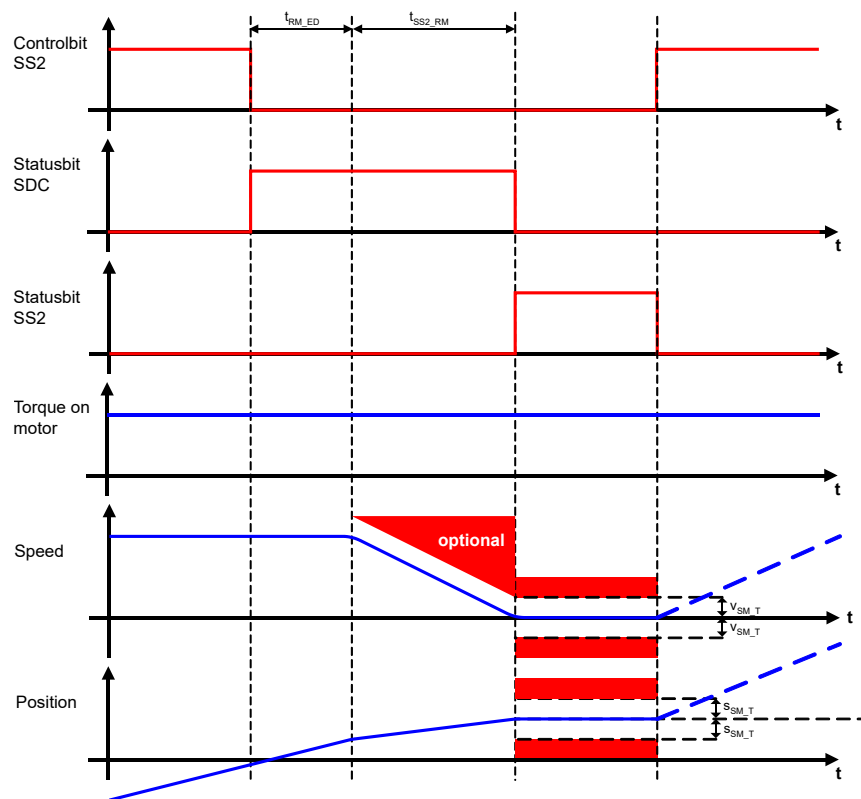


Figure 48: Safe Stop 2

The monitored ramp always starts at the currently monitored limit and is calculated using the parameterized slope. If no speed limit is specified, the monitored ramp starts at the **maximum possible speed $2^{31} - 1 = 2147483647$ [unit/s]**. This does not make sense in normal cases! In order to start the ramp at a physically reasonable value, it is recommended to limit the currently monitored speed limit using safety function Safe Maximum Speed (SMS), for example.

Application of the safety function

After being stopped by the standard application, the drive must be actively held at standstill.

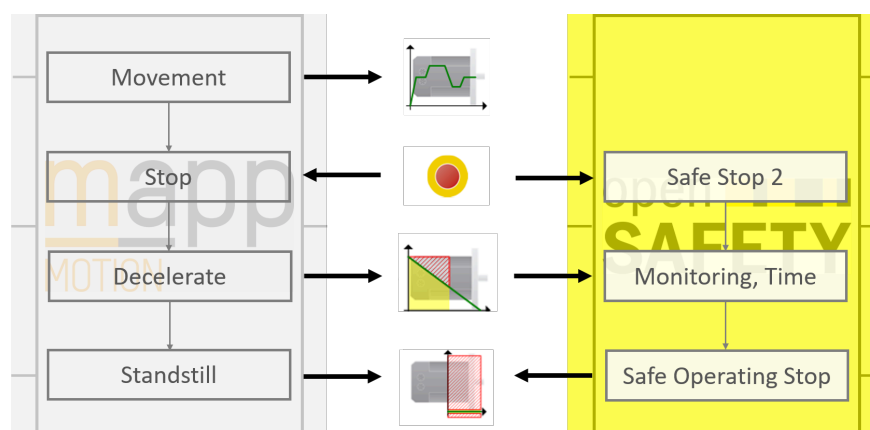


Figure 49: Safe Stop 2



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \ Integrated safety functions \ Safe Stop 2, SS2

Exercise: Safe standstill after movement (SS2)

The goal of this exercise is to evaluate the manual and automatic mode and control SS2.

Automation Studio:

- 1) Create variables "gdiManual" and "gdiAutomatic" and connect them to inputs.
- 2) Implement the automatic mode → Movement with constant speed, light curtain is active.
- 3) Implement the manual mode → Movement when button is pressed, light curtain is not active.
- 4) Compiling

SafeDESIGNER:

- 1) Evaluate the switch position (manual and automatic mode).
- 2) Change "Pulse source" at SafeDigitalInput04 to "Pulse 3".
- 3) SS2 is enabled in manual mode 1 second after the green button is released.
- 4) Compiling

Transferring the SafeAPPLICATION**Testing:**

- 1) Enable the manual mode.
- 2) Move the axis by pressing the green button.
- 3) Trigger the light curtain → The axis must continue to rotate.
- 4) Release the green button → The axis must stop but the controller remains switched on.
- 5) SS2 is active for one second after the green button has been released.
- 6) Move the axis in monitor mode with "MoveAbsolute" to trigger safety violations.
- 7) Check DriveEnable of output mpAxisBasic and the axis status on the ACOPOS display module.

```

IF ((gdiEStop = FALSE) OR
    (gdiLightcurtain = FALSE AND gdiAutomatic = TRUE)) THEN
    MpAxisBasic_0.Power := FALSE;
    sStep := enERROR;
END_IF;

```

Table 4: Example code for the light curtain and automatic mode - Automation Studio

```

IF (gdiManual = TRUE) THEN
    MpAxisBasic_0.MoveVelocity := FALSE;
    MpAxisBasic_0.JogPositive := gdiReset;
ELSIF (gdiAutomatic = TRUE) THEN
    MpAxisBasic_0.MoveVelocity := TRUE;
ELSE
    MpAxisBasic_0.MoveVelocity := FALSE;
END_IF;

IF (MpAxisBasic_0.Error = TRUE) THEN
    sStep := enERROR;
END_IF

```

Table 5: Example code for the manual and automatic mode - Automation Studio

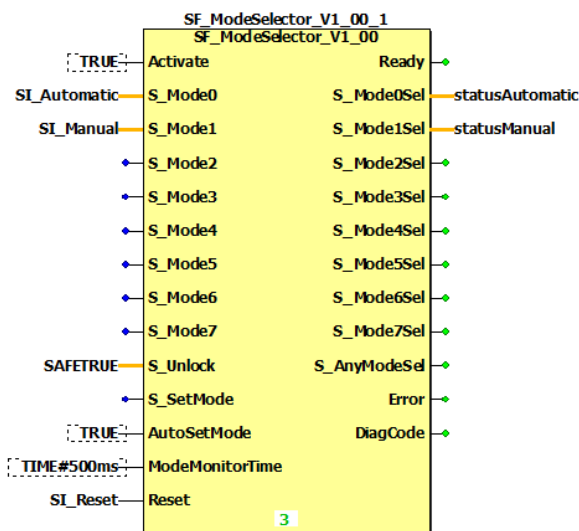


Figure 50: Example code for function block ModeSelector - SafeDESIGNER

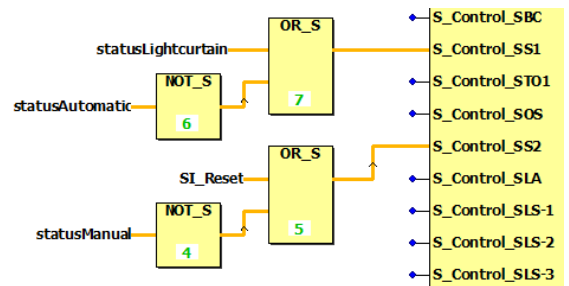


Figure 51: Example code for the connection of SS1 and SS2 - SafeDESIGNER

Parameter	Value	Unit
Axis01 - Speed functions - SS2		
SS2 - Ramp monitoring - Enabl	Enabled	
SS2 - Ramp monitoring - Time	1000000	us

Figure 52: Parameter SS2 - SafeDESIGNER

5.5 Safely Limited Speed (SLS)

Parameter "**Ramp monitoring - Enable delay time**" defines the delay time between requesting and starting the monitoring.

The SLS safety function is used to monitor up to 4 specified speed limits "**SLS1/2/3/4 - Speed limit**" and, in addition to the time, ramp "**SLS - Ramp monitoring**" when the SLS function is requested.

All limits can be monitored in parallel. If a request is made to monitor multiple speed limits at the same time, the lowest limit value is always monitored. To make this possible, the function block includes four different inputs **S_RequestSLSX-1/2/3/4**.

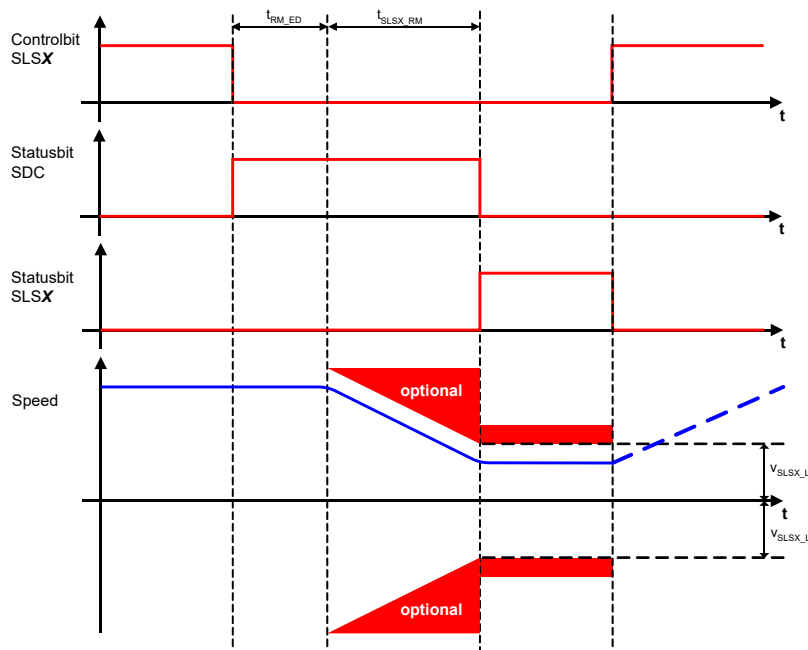


Figure 53: Safe Limited Speed

Application of the safety function

Maintenance tasks and setup mode often require work to be performed on a machine while it is running. This increases the potential for hazardous situations. SLS monitors whether the drive speed is below a limit value. This ensures greater safety when working on the machine while it is running.

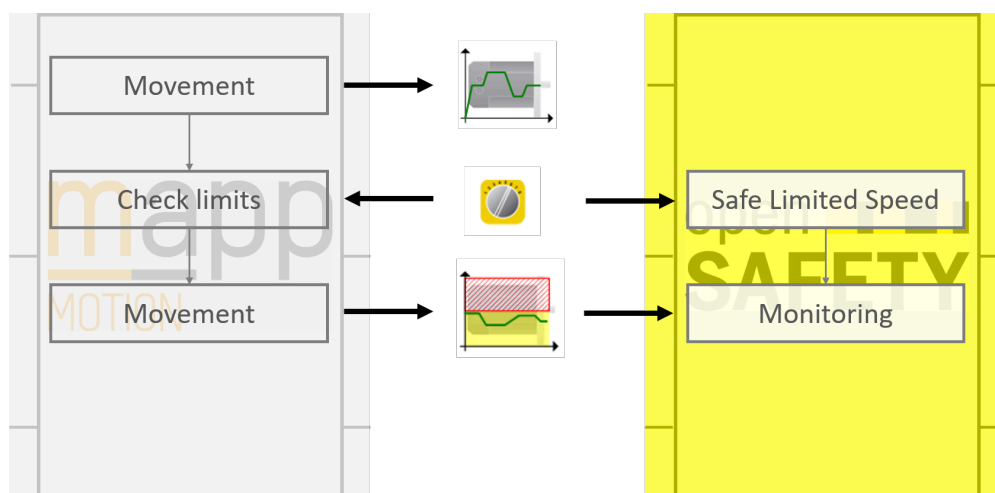


Figure 54: Safe Limited Speed



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \ Integrated safety functions \ Safely Limited Speed, SLS

Safe Maximum Speed (SMS)

The main difference between safety function Safe Maximum Speed and SLS is that it cannot be actively requested. It is either enabled (parameter "SMS - Enable = Enabled") or disabled (parameter "SMS - Enable = Disabled") in the configuration.

When enabled, the current speed is constantly monitored according to a defined limit (parameter "SMS - Speed limit").



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \ Integrated safety functions \ Safe Maximum Speed, SMS

Exercise: Manual mode limits speed

The goal of this exercise is to define the maximum speed and to enable a safely limited speed of 500 units/s when the light curtain is triggered during a movement in manual mode.

SafeDESIGNER:

- 1) Set "Ramp monitoring - Enable delay time" with 100 milliseconds.
- 2) Enable "SMS - Enable".
- 3) Set "SMS - Speed limit" to 5100 units/s.
- 4) Set "SLS1 - Speed limit" to 550 units/s.
- 5) Set "SLS1 - Ramp monitoring - Time" to 1 second.
- 6) Connect the signal from the light curtain, manual mode and the green button with input "S_Control_SLS1".
- 7) Compiling

Automation Studio:

- 1) Decrease speed "parAxis.Jog.Velocity" when enabling the light curtain.
- 2) Update the parameter at runtime using "MpAxisBasic_0".
- 3) Compiling

Transferring the SafeAPPLICATION

Testing:

- 1) Start a manual movement.
- 2) Trigger the light curtain and observe the changed movement.
- 3) Check output DriveEnable of mpAxisBasic and the axis status on the ACOPOS display module.

```

IF (gdiManual := TRUE) THEN
    MpAxisBasic_0.MoveVelocity := FALSE;
    MpAxisBasic_0.JogPositive := gdiReset;

    IF (gdiLightcurtain = TRUE) THEN
        parAxis.Jog.Velocity := 2;
        MpAxisBasic_0.Update := TRUE;
        MpAxisBasic_0();
    ELSE
        parAxis.Jog.Velocity := 0.5;
        MpAxisBasic_0.Update := TRUE;
        MpAxisBasic_0();
    END_IF;
END_IF;

```

Table 6: Example code for reducing the speed - Automation Studio

```

MpAxisBasic_0.Update := FALSE;
MpAxisBasic_0.ErrorReset := gdiReset;
MpAxisBasic_0();

```

Table 7: Example code for resetting the update command - Automation Studio

Parameter	Value	Unit
Axis01 - Speed functions - SMS/SLS		
SMS - Enable	Enabled	
SMS - Speed limit	5200	units/s
SLS1 - Speed limit	500	units/s
SLS2 - Speed limit	0	units/s
SLS3 - Speed limit	0	units/s
SLS4 - Speed limit	0	units/s
SLS - Ramp monitoring - Enable	Enabled	
SLS1 - Ramp monitoring - Time	1000000	us
SLS2 - Ramp monitoring - Time	0	us
SLS3 - Ramp monitoring - Time	0	us
SLS4 - Ramp monitoring - Time	0	us

Figure 55: Parameter SLS1 - SafeDESIGNER

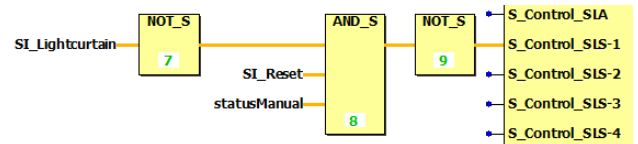


Figure 56: Example code for wiring SLS1 - SafeDESIGNER

5.6 Safely Limited Increment (SLI)

With the SLI safety function, a movement is monitored with regard to a defined position limit.

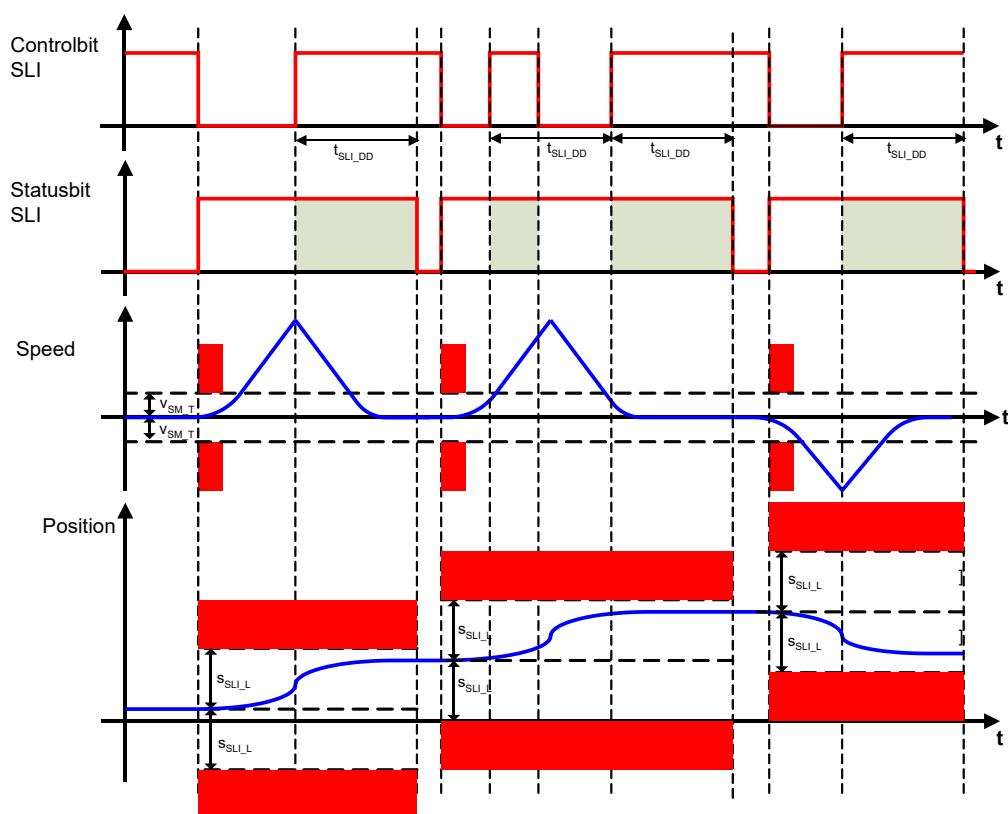


Figure 57: Safe Limited Speed

The safe axis must be at standstill when this function is enabled. A position window that is safety-monitored is then generated. This position window depends on the parameterized safe position limit "SLI - Position limit".

The standard application must guarantee that this position window is not exceeded. The evaluation is performed via the safe encoder position. Mechanical moving or bumping can thus also lead to a safety violation. Safe pulse disabling is enabled immediately and an acknowledgeable error state is triggered.

After the safety function is disabled, monitoring continues for the configured period of time "SLI - Disable delay time". This prevents continuous movement caused by constant jogging.



Safety technology \ mapp Safety \ Engineering \ SafeMOTION \ Integrated safety functions \ Safely Limited Increment, SLI

Exercise: SLI in manual mode

The goal of this exercise is to be permitted to move the axis for a maximum of 20 revolutions in manual mode.

SafeDESIGNER:

- 1) Connect the inverted status of the manual mode and the green button to SLI.
- 2) Parameterize "SLI - Position limit" to 20 revolutions.
- 3) Limit "SLI - Disable delay time" to 2 seconds.
- 4) Building and transferring

Transferring the SafeAPPLICATION

Testing:

- 1) Start a manual movement.
- 2) Check output DriveEnable of mpAxisBasic and the axis status on the ACOPOS display module.

Parameter	Value	Unit
Axis01 - Advanced functions - SLI		
SLI - Position limit	20000	units
SLI - Disable delay ti	2000000	us

Figure 58: Parameter SLI - SafeDESIGNER

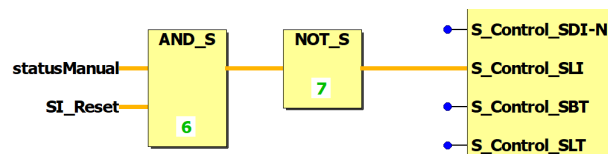


Figure 59: Example code for SLI - SafeDESIGNER

Exercise: Evaluate limit values in mapp Cockpit

The goal of this exercise is to record and diagnose the safe speed and the change of the limit values in mapp Cockpit.

Automation Studio:

- 1) Add the mapp Cockpit configuration and assign role "Everyone".
- 2) Building and transferring

Google Chrome:

- 1) Open mapp Cockpit (IP address:8084/mappCockpit).
- 2) Configure a trace.
 - *ACP:IF3.ST3_Axis1:196 (ParId 196 - Safe speed)
 - *ACP:IF3.ST3_Axis1:224 (ParId 224 - Speed limit value)
- 3) Start a manual movement.
- 4) Evaluate safe speed and limit values.



Diagnostics and service / Diagnostics tools / Getting started with mapp Cockpit

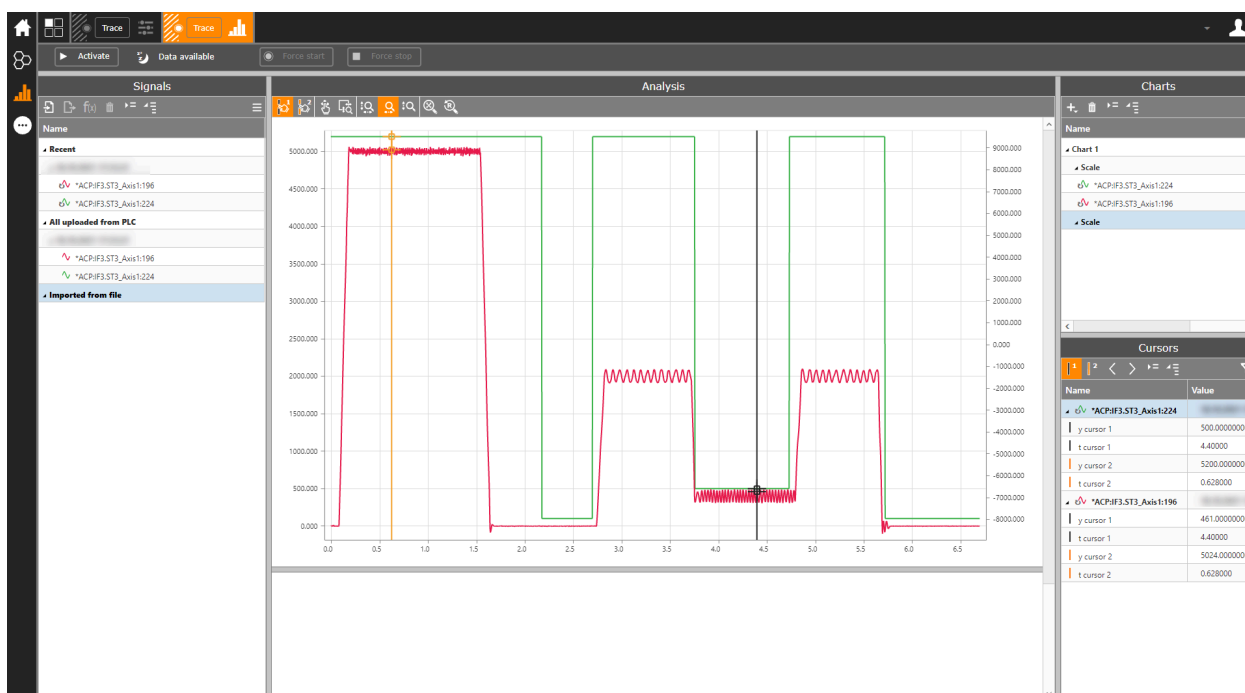


Figure 60: Record safe speed and limit values of the axis in mapp Cockpit

5.7 Other SafeMOTION functions

This training module contains an overview of some integrated safety functions. For additional information about the supported safety functions, see Automation Help:



Hardware \ Motion control \ SafeMOTION \ Safety technology \ Integrated safety functions

6 Example projects and solutions



The example code in this documentation is exclusively for implementing the test setup for this training. This example code is not permitted to be adopted for safety applications under any circumstances. Each safe machine application must be subjected to a separate risk analysis and a safety concept must be created.

6.1 Example solution - SafeDESIGNER project

Emergency stop and light curtain

Using function blocks "SF_EmergencyStop_V1" and "SF_ESPE_V1", the signals of the two sensors are evaluated and acknowledged with the green button.

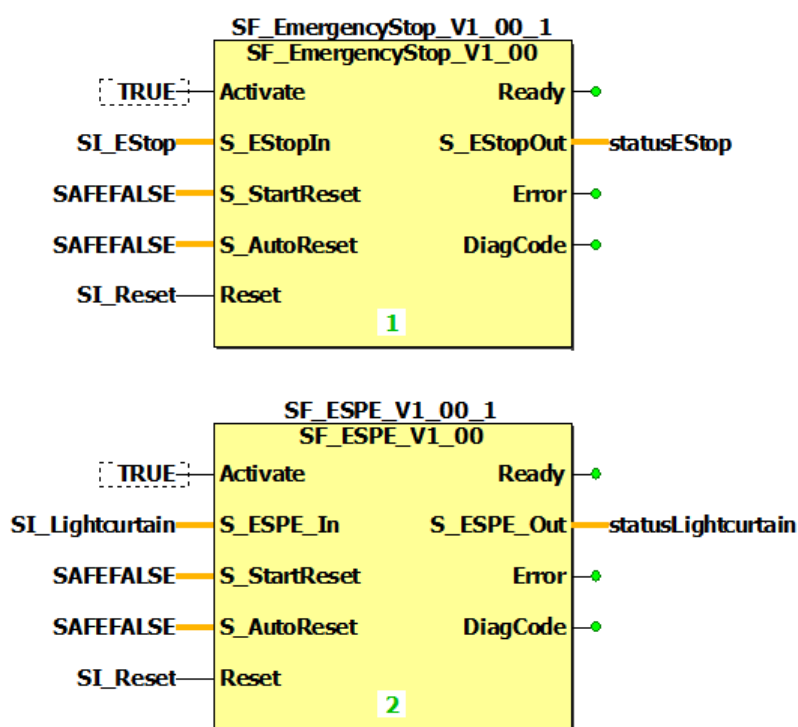


Figure 61: Emergency stop and light grid evaluation

Mode selector switch

Function block "SF_ModeSelector_V1" is used to evaluate the manual mode and the automatic mode.

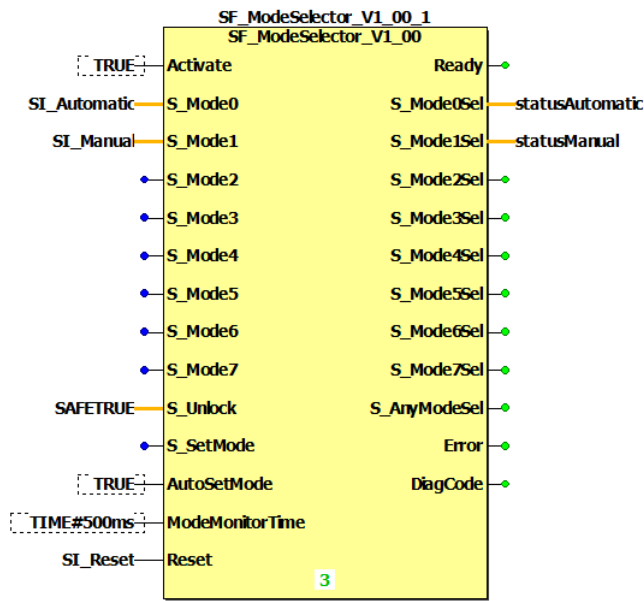


Figure 62: ModeSelector block

Function block: Safety_Function

PLCopen function block "SF_oS_MOTION_BR" is used for the active monitoring of four different parameters, such as speed, position or acceleration.

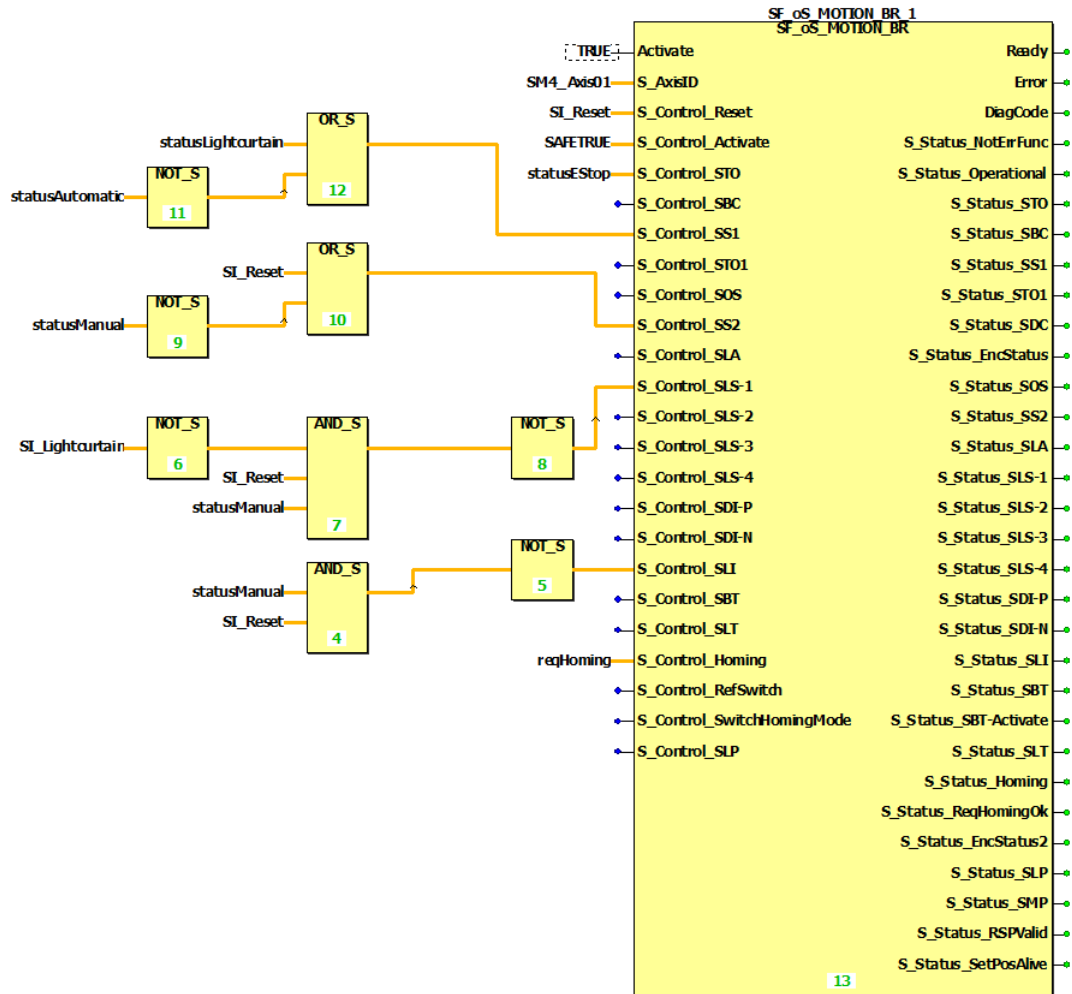


Figure 63: SafeMOTION function block

6.2 Example solution - Automation Studio project

Init section

In the Init section, input ".Enable" of the mpAxisBasic block is enabled.

```
MpAxisBasic_0.Enable := TRUE;
```

Cyclic section

The status of the emergency stop function and light curtain is queried first and the axis is switched off. A step switching chain is then used to automatically switch on the axis, home it and move it according to the set mode. Error handling is also part of this code.

```
MpAxisBasic_0.Paramters := ADR(parAxis);
MpAxisBasic_0.MpLink := ADR(gAxis_1);

IF (gdiEStop = FALSE OR gdiLightcurtain = FALSE
AND gdiAutomatic = TRUE) THEN
    MpAxisBasic_0.Power := FALSE;
    sStep := enERROR;
END_IF;

CASE sStep OF
    enINIT:
        IF (gdiReset = TRUE) THEN
            sStep := enSTART;
        END_IF;
```

```

enSTART:
  IF (MpAxisBasic_0.Info.ReadyToPowerOn = TRUE
    AND MpAxisBasic_0.Active = TRUE) THEN
    sStep := enPOWERON;
  END_IF;

enPOWERON:
  MpAxisBasic_0.Power := TRUE;
  IF (MpAxisBasic_0.PowerOn = TRUE) THEN
    sStep := enHOME;
  END_IF;

enHOME:
  MpAxisBasic_0.Home := TRUE;
  IF (MpAxisBasic_0.IsHomed = TRUE) THEN
    MpAxisBasic_0.Home := FALSE;
    sStep := enOPERATION;
  END_IF;

enOPERATION:
  IF (gdiManual = TRUE) THEN
    MpAxisBasic_0.MoveVelocity := FALSE;
    MpAxisBasic_0.JogPositive := gdiReset;

    IF (gdiLightcurtain = TRUE) THEN
      parAxis.Jog.Velocity := 2;
      MpAxisBasic_0.Update := TRUE;
      MpAxisBasic_0();
    ELSE
      parAxis.Jog.Velocity := 0.4;
      MpAxisBasic_0.Update := TRUE;
      MpAxisBasic_0();
    END_IF

  ELSIF (gdiAutomatic = TRUE) THEN
    MpAxisBasic_0.MoveVelocity := TRUE;

  ELSE
    MpAxisBasic_0.MoveVelocity := FALSE;
  END_IF

  IF (MpAxisBasic_0.Error = TRUE) THEN
    sStep := enERROR;
  END_IF

enERROR:
  MpAxisBasic_0.MoveVelocity := FALSE;
  IF (MpAxisBasic_0.Error = FALSE
    AND MpAxisBasic_0.StatusID = 0) THEN
    IF (MpAxisBasic_0.PowerOn = FALSE
      AND MpAxisBasic_0.Power = TRUE) THEN
      MpAxisBasic_0.Power := FALSE;
    END_IF;
    sStep := enINIT;
  END_IF
END_CASE;

MpAxisBasic_0.Update := FALSE;
MpAxisBasic_0.ErrorReset := gdiReset;
MpAxisBasic_0();

```

6.3 Sample solution - Variables

6.3.1 SafeDESIGNER

Global variables

This file contains all variables that are connected to a safe input or output as well as the variables for the communication channels.

	Name	Data type	Description	Terminal	Init
1	SafeInputs				
2	SM4_Axis01	SAFEINT			SAFEINT#0
3	SI_Reset	SAFEBOOL		SL1.SM3.SafeDigitalInput05	
4	SI_EStop	SAFEBOOL		SL1.SM3.SafeTwoChannelInput0102	
5	SI_Lightcurtain	SAFEBOOL		SL1.SM2.SafeTwoChannelInput0102	
6	SI_Automatic	SAFEBOOL		SL1.SM3.SafeDigitalInput03	
7	SI_Manual	SAFEBOOL		SL1.SM3.SafeDigitalInput04	

Figure 64: Global variables - SafeDESIGNER

Local variables: Main code

The function blocks and the status variables from the evaluation of the emergency stop function, light curtain and mode selector switch are declared here.

	Name	Data type	Usage
1	FUBs		
2	SF_EmergencyStop_V1...	SF_EmergencySto...	VAR
3	SF_ESPE_V1_00_1	SF_ESPE_V1_00	VAR
4	SF_ModeSelector_V1...	SF_ModeSelector...	VAR
5	SF_oS_MOTION_BR_1	SF_oS_MOTION_BR	VAR
6	Request variables		
7	reqHoming	SAFEBOOL	VAR
8	Status variables		
9	statusEStop	SAFEBOOL	VAR
10	statusLightcurtain	SAFEBOOL	VAR
11	statusAutomatic	SAFEBOOL	VAR
12	statusManual	SAFEBOOL	VAR

Figure 65: Local variable declaration for main code

6.3.2 Automation Studio

Global variables

The global variable file contains the variables that are directly connected to the inputs.

Name	Type
gdiEStop	BOOL
gdiReset	BOOL
gdiLightcurtain	BOOL
gdiAutomatic	BOOL
gdiManual	BOOL

```
VAR
    gdiEStop : BOOL;
    gdiReset : BOOL;
    gdiLightcurtain : BOOL;
    gdiAutomatic : BOOL;
    gdiManual : BOOL;
END_VAR
```

Text view of the global variables

Figure 66: Global variables - Automation Studio

Local variables and types

The local variable file contains the instance of the MpAxisBasic block, the associated parameter structure and the variable for the step switching chain. The individual steps are created as an enumeration in the type file. This is done to improve readability of the code. Simple numerical values can also be used instead of this enumeration.

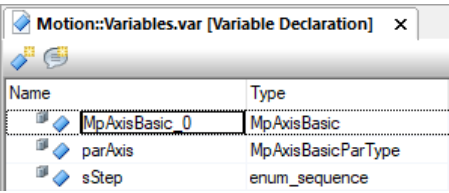


Figure 67: Local variables - Automation Studio

```
VAR
    MpAxisBasic_0 : MpAxisBasic;
    parAxis : MpAxisBasicParType;
    sStep : enum_sequence;
END_VAR
```

Text view of the local variables

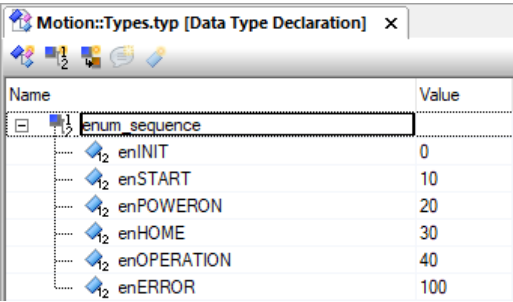


Figure 68: Local types - Automation Studio

```
TYPE
    enum_sequence :
        ( enINIT := 0,
          enSTART := 10,
          enPOWERON := 20,
          enHOME := 30,
          enOPERATION := 40,
          enERROR := 100);
END_TYPE
```

Text view of the local types

6.4 Sample solution - Parameters for the safe hardware modules

Safe input module X20SI2100

The inputs from the safe input module are wired to the light curtain. This sensor already has its own test pulse, so it has to be disabled on the module and a filter must be configured. The light curtain is evaluated via multi-channel evaluation using an equivalent signal.

Order no.:	X20SI2100	
Description:	X20 Safe Digital In, 2x1, 24V	
SafeNODE ID:	2	
Import file:	-	
Parameter	Value	Unit
SafeDigitalInput01		
Pulse source	No pulse	
Filter off	1000	us
Filter on	200000	us
Discrepancy time	50000	us
Dual-channel processing mode	Equivalent	
SafeDigitalInput02		
Pulse source	No pulse	
Filter off	1000	us
Filter on	200000	us

Figure 69: Parameter X20SL8101

Safe mixed module X20SC0806

The safe mixed module is wired to the emergency stop, the mode selector switch, the green button and the "Saw" output. The emergency stop signal is read out equivalently. The pulse for the mode selector switch and green button must be changed.

Order no.:	X20SC0806	
Description:	X20 Safe Combined Module, 8x1 24V, 6x1 24V 0.2A	
SafeNODE ID:	3	
Import file:	-	

Parameter	Value	Unit
SafeDigitalInput01		
Pulse source	Pulse 1	
Filter off	0	us
Filter on	200000	us
Discrepancy time	50000	us
Dual-channel processing mode	Equivalent	
SafeDigitalInput02		
Pulse source	Pulse 2	
Filter off	0	us
Filter on	200000	us
SafeDigitalInput03		
Pulse source	Pulse 3	
Filter off	0	us
Filter on	200000	us
Discrepancy time	50000	us
Dual-channel processing mode	Equivalent	
SafeDigitalInput04		
Pulse source	Pulse 3	
Filter off	0	us
Filter on	200000	us
SafeDigitalInput05		
Pulse source	Pulse 4	
Filter off	0	us
Filter on	200000	us
Discrepancy time	50000	us
Dual-channel processing mode	Equivalent	

Figure 70: Parameter X20SC0806

Safe axis module 8ESMC59315

The safe axis must be parameterized with the correct values depending on the safety function used. The screenshot shows which parameters are necessary to run the sample application correctly.

Axis01 - General settings - Standstill monitoring		
Standstill monitoring - Position tolerance	5	units
Standstill monitoring - Speed tolerance	100	units/s
Axis01 - General settings - Ramp monitoring		
Ramp monitoring - Speed deceleration limit	1073676289	units/s ²
Ramp monitoring - Enable delay time	100000	us
Axis01 - Basic functions - SS1		
SS1 - Ramp monitoring - Enable	Enabled	
SS1 - Ramp monitoring - Time	3000000	us
Axis01 - Speed functions - SS2		
SS2 - Ramp monitoring - Enable	Enabled	
SS2 - Ramp monitoring - Time	1000000	us
Axis01 - Speed functions - SMS/SLS		
SMS - Enable	Enabled	
SMS - Speed limit	5200	units/s
SLS1 - Speed limit	500	units/s
SLS2 - Speed limit	0	units/s
SLS3 - Speed limit	0	units/s
SLS4 - Speed limit	0	units/s
SLS - Ramp monitoring - Enable	Enabled	
SLS1 - Ramp monitoring - Time	1000000	us
SLS2 - Ramp monitoring - Time	0	us
SLS3 - Ramp monitoring - Time	0	us
SLS4 - Ramp monitoring - Time	0	us
Axis01 - Advanced functions - SLI		
SLI - Position limit	20000	units
SLI - Disable delay time	2000000	us

Figure 71: Parameter 8ESMC59315

7 Summary

With safe drive technology SafeMOTION, the drive is controlled via the PLC and only monitored via the safety functions on the SafeLOGIC controller.

The hardware itself is configured in Automation Studio and commissioned using mapp Motion, mapp Safety and mapp Cockpit.

SafeDESIGNER uses PLCopen blocks to evaluate sensors such as the light curtain or emergency stop function and to monitor various movements. However, this application does not actively intervene in the control process of motion control, but enables safe pulse disabling and disables the holding brake output in the case of a safety violation.



Figure 72: mapp Safety offers the full range of functions for the entire hardware portfolio

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